Principled Remasking in Masked Diffusion Language Models: A Comparative Literature Review

MSc Thesis — Bocconi University, supervised by Prof. Giacomo Zanella

March 2026

Contents

1 Introduction

This document is a self-contained literature review for the MSc thesis *"Principled Remasking in Masked Diffusion Language Models"* (Bocconi University, 2026). It covers twelve papers organised into five parts that follow the intellectual development of the field from its earliest precursors to the state of the art.

The central question of the thesis. Masked Diffusion Models (MDMs) generate text by iteratively unmasking tokens from a fully masked sequence. When multiple tokens are unmasked simultaneously — necessary for efficiency — the model treats them as conditionally independent, introducing a *factorization error* E_{fact}. Remasking (re-masking a committed token and re-predicting it in richer context) is a natural corrective, but no prior work has characterised it within a rigorous error-bound framework. The thesis asks:

> *Can we design principled, theoretically grounded remasking strategies that reduce E_{fact} without retraining the model, and derive conditions under which training-free confidence signals are sufficient?*

How to read this document. The five parts are ordered to build understanding progressively:

- **Part I** situates the thesis in the pre-diffusion literature (Mask-Predict, 2019), showing that iterative confidence-guided remasking has been empirically effective for years — but has never had a rigorous foundation.
- **Part II** builds the discrete diffusion framework from scratch: D3PM (foundational theory), SEDD (score-based alternative with PC samplers), MDLM (practical simplification), and LLaDA (8B-scale validation).
- **Part III** introduces the Discrete Flow Matching perspective, which provides a clean continuous-time language for corrector steps.
- **Part IV** covers the two papers most directly foundational to the thesis: Lavenant & Zanella (KL error decomposition) and the EB-Sampler (optimal unmasking schedule). The thesis extends these to the remasking direction.
- **Part V** surveys all existing remasking approaches: ReMDM, Informed Correctors, RemeDi, and PRISM — from the most theoretical to the most empirically powerful.
- **Part VI** is a structured cross-paper comparison on eight dimensions.

Each paper is reviewed under six headings: (A) Summary, (B) Method Details, (C) Theoretical Contributions, (D) Confidence Signal, (E) Limitations, (F) Relation to This Thesis.

2 How to Use This Document

2.1 Reading Roadmap

This document covers twelve papers in five thematic parts. The intellectual dependency graph is not linear — read in the order below to build concepts progressively:

```
STEP 1 - Foundation (what problem are we solving?):
  MDLM (Part II) → understand the forward/reverse process and the training loss
  D3PM  (Part II) → understand where the independence assumption comes from

STEP 2 - The core error (what exactly goes wrong?):
  Lavenant & Zanella (Part IV) → definition of E_fact; the information profile I(x)
  EB-Sampler (Part IV)         → how to minimise E_fact for unmasking

STEP 3 - Remasking approaches (how do we fix committed errors?):
  ReMDM (Part V)               → principled remasking posterior  _t
  Informed Correctors (Part V) → MCMC theory; why confidence-guided > uniform
  PRISM (Part V)               → learned confidence with formal guarantee
  RemeDi (Part V)              → RL-finetuned policy (upper bound)

STEP 4 - Theoretical unification:
  DFM (Part III)               → correctors as Gibbs samplers; velocity field
  SEDD (Part II)               → score-based view; PC corrector

STEP 5 - Scale and history:
  LLaDA (Part II)              → MDMs at 8B scale; why uniform schedule is wasteful
  Mask-Predict (Part I)        → heuristic ancestor; what the thesis formalises
```

The central question to keep in mind throughout: > *Each paper that does remasking has a different answer to: "which tokens should be > remasked, and why?" Does the choice have a formal justification?*

2.2 Six Questions That Thread Through the Document

These questions connect all twelve papers. As you read each section, ask:

1. **What is E_{fact} for this algorithm?** Some papers reduce it explicitly (EB-Sampler, thesis); others reduce it implicitly (ReMDM, Informed Correctors); others ignore it entirely (D3PM, LLaDA).

2. **What confidence signal does this paper use?** Map it to Tier 1/2/3. Tier 1 requires $O(L)$ forward passes; Tier 2 requires training; Tier 3 is free from logits.

3. **Does this paper require training, retraining, or nothing?** The training axis is the primary practical dimension that distinguishes the approaches.

4. **What is the paper's relation to $I(x)$?** The information profile is the theoretical gold standard; papers that ignore it leave E_{fact} on the table.

5. **What does the paper's confidence signal converge to as $p_\theta \to q$?** At the optimal model, all Tier 3 signals should equal $I^i(x)$. Do they?

6. **What is the compute cost per generated token?** Expressed as number of forward evaluations (NFE) and how it scales with sequence length and correction count.

2.3 Unified Notation Reference

All twelve papers use overlapping but inconsistent notation. This table fixes a single notation used consistently throughout this document.

Symbol	Meaning	Defined in
L	Sequence length	—
$\mathcal{V}$	Token vocabulary (size V)	—
$x = (x^1, \dots, x^L) \in \mathcal{V}^L$	Clean token sequence	D3PM
$m \notin \mathcal{V}$	Mask token	D3PM
$z_t \in (\mathcal{V} \cup \{m\})^L$	Noisy sequence at noise level t	D3PM
$\mathcal{M}_t = \{i : z_t^i = m\}$	Set of masked positions at time t	MDLM
T	Total number of denoising steps	D3PM
$\alpha_t \in [0, 1]$	Survival probability: $P(z_t^i = x^i)$; $\alpha_T \approx 0$, $\alpha_0 = 1$	MDLM
$\beta_t = 1 - \alpha_t / \alpha_{t-1}$	Per-step masking rate	D3PM
$q(z_t \mid x) = \prod_i \mathrm{Cat}(z_t^i; \alpha_t e_{x^i} + (1 - \alpha_t) e_m)$	Forward kernel (absorbing)	D3PM / MDLM
$p_\theta(x^i \mid z_t)$	Model's predicted clean-token distribution at position i	MDLM
$p_\theta(z_{t-1}^i \mid z_t)$	Reverse kernel for position i	MDLM
π	True data distribution	Lav. & Zan.
p_{alg}	Distribution of algorithm's output sequences	Lav. & Zan.
$I^i(x) = H(x^i \mid x^{\ i})$	True conditional entropy of position i (info. profile)	Lav. & Zan.
$I(x) = \sum_i I^i(x)$	Total sequence information	Lav. & Zan.
$\hat{I}^i(z_t) = H(p_\theta(x^i \mid z_t))$	Model's entropy estimate at position i	EB-Sampler
$\Sigma^2 = \mathrm{Var}_i[I^i(x)]$	Variance of the information profile	Lav. & Zan.
E_{fact}	Factorization error: KL due to independence assumption	Lav. & Zan.
E_{learn}	Learning error: KL due to imperfect p_θ	Lav. & Zan.
ε	Entropy budget per step (EB-Sampler)	EB-Sampler

Symbol	Meaning	Defined in
$\sigma_t^i = 1 - \alpha_t$	ReMDM remasking probability for committed token at time t	ReMDM
R	Number of corrector passes per denoising step	ReMDM
τ	Remasking threshold (remask if $\hat{I}^i > \tau$)	Thesis
$c^i_{\mathrm{mp}} = \max_j p_\theta^{ij}$	Max-probability confidence (Tier 3)	Mask-Predict
$c^i_H = -H(p_\theta(x^i \mid z_t))$	Entropy confidence (Tier 3; $= -\hat{I}^i$)	ReMDM / EB-Sampler
$c^i_{\mathrm{mg}} = p_\theta^{(1)} - p_\theta^{(2)}$	Margin confidence (Tier 3)	—
$g^i_\phi \in [0,1]$	PRISM quality head output (Tier 2)	PRISM
$\psi^i \in [0,1]$	RemeDi UPS output (Tier 2)	RemeDi
$s_\theta(z_t)_{ij} \approx p_t(z_t^{(j)})/p_t(z_t)$	SEDD concrete score at position i	SEDD
$u_t(z^i = m \to j)$	DFM velocity field for position i	DFM
λ	Spectral gap of Gibbs corrector chain	Inf. Corr.
$\mathcal{L}_{\mathrm{MDM}}$	MDLM training objective (weighted masked cross-entropy)	MDLM

Key relations to memorise:

$$\hat{I}^i(z_t) \xrightarrow{p_\theta \to q} I^i(x) \quad \text{(at optimality, entropy estimate = true info)}$$

$$\sigma_t^i = 1 - \alpha_t \quad \text{(ReMDM: remask each committed token with this prob.)}$$

$$E_{\mathrm{fact}} \le C \cdot \frac{\Sigma^2}{T} \quad \text{(fewer steps or higher profile variance} \to \text{larger error)}$$

$$E_{\mathrm{fact}}(\mathrm{EB}) \le \varepsilon \cdot \max_i \hat{I}^i \quad \text{(EB-Sampler bound)}$$

3 Part I — Before Diffusion: Non-Autoregressive Iterative Decoding

---

3.1 Mask-Predict (2019) — The Heuristic Ancestor

Full citation: Ghazvininejad, Levy, Liu & Zettlemoyer (2019), "Mask-Predict: Parallel Decoding of Conditional Masked Language Models." *EMNLP 2019.* arXiv:1904.09324.

3.1.1 Summary

Mask-Predict is the earliest paper to demonstrate confidence-guided iterative remasking for language generation, in the context of non-autoregressive (NAR) machine translation. Prior to this work, NAR models generated all target tokens in parallel in a single pass, achieving high speed at the cost of quality (due to multi-modality in the output distribution). Mask-Predict proposes a simple iterative refinement: starting from a fully masked target, alternate between predicting all masked tokens and re-masking the lowest-confidence predictions. This converges in $O(T)$ parallel steps — 4–9× faster than autoregressive decoding — with BLEU scores competitive with strong autoregressive baselines on WMT EN↔DE and EN↔ZH.

The paper's contribution is purely algorithmic. It does not provide a probabilistic model of the forward/reverse process, a noise schedule, or any theoretical justification for why remasking should help. It is effective because of the simple intuition that a token predicted in a context where all other tokens are unmasked is more likely to be correct than one predicted when most of the sequence is still masked. This intuition is exactly what the thesis formalises: the information profile $I(x)$ captures how much each position benefits from additional context, and remasking targets the positions with the most to gain.

3.1.2 Method Details

The model is a BERT-style bidirectional transformer trained with masked language model objectives on parallel translation data. Given a source sentence x and a target length n, Mask-Predict initialises $y = [\mathrm{M}]^n$ and iterates:

Step 1 — Predict. Run the masked LM conditioned on x and the current y:

$$p^i = P(y^i \mid x,\ y^{\ i}) \quad \forall i : y^i = [\mathrm{M}]$$

For unmasked positions, the probability of the current token is used directly.

Step 2 — Re-mask. Compute the confidence of every position as $c^i = \max_j p^i_j$ (max-probability). Re-mask the n_t lowest-confidence positions:

$$n_t = \left\lceil n \cdot \frac{T - t}{T} \right\rceil$$

so that n_t decreases linearly from n to 0 over T iterations.

Full algorithm:

```
y = [M, M, ..., M]
for t = 1, ..., T:
    # Predict all masked positions in parallel
    for i with y^i = [M]:
        p^i = P(y^i | x, y\{y^i=[M]})
    # Commit all positions
    y^i ← argmax p^i   for all i
    # Remask n_t lowest-confidence committed tokens
    n_t = ceil(n * (T - t) / T)
    remask_set ← argsort(c)[:n_t]   # lowest confidence first
    y^i ← [M]   for i in remask_set
return y
```

The confidence signal is thus $c^i = \max_j p_\theta(y^i = j \mid x, y^{\;i})$, the simplest possible measure of how certain the model is about position i.

Masking schedule. The number of masked tokens decreases linearly: $n_t/n = (T - t)/T$. This is the *linear schedule*, one of the schedules implemented in the thesis's `schedule` strategy.

3.1.3 Theoretical Contributions

None. Mask-Predict is purely empirical. The authors justify the approach informally: low-confidence tokens are likely wrong, and re-masking them allows the model to predict them in a richer context (once more tokens are committed). There is no probabilistic model, no noise schedule, no ELBO, and no convergence guarantee.

3.1.4 Confidence Signal

$c^i = \max_j p_\theta(y^i = j \mid x, y^{\;i})$ — the *max-probability* of the top-1 prediction at position i.

Properties: - Cheap: computed from the standard forward-pass logits at no extra cost. - Heuristic: no formal guarantee that low c^i implies high prediction error. - Miscalibrated across noise levels: the same value of c^i means different things depending on how many other tokens are masked.

This is the Tier 3 (heuristic) confidence signal in the thesis's three-tier hierarchy. The thesis asks formally: under what conditions is c^i a consistent estimator of $H(x^i \mid \text{context})$?

3.1.5 Limitations

1. **No probabilistic framework.** Mask-Predict is not a generative model; it is a deterministic iterative decoding algorithm applied to a trained masked LM. There is no forward process, no reverse process, and no connection to diffusion.
2. **Conditional generation only.** The model requires a source sentence x (machine translation); it is not directly applicable to unconditional or prompted language generation.
3. **No convergence guarantee.** There is no proof that the iterative procedure converges, that it samples from the right distribution, or that more iterations improve quality.
4. **Miscalibrated confidence.** Using max-probability as a confidence signal conflates model confidence with prediction accuracy, and ignores the dependence on the current masking rate.
5. **Length must be pre-specified.** The target length n is sampled from a prior; incorrect length predictions cannot be corrected.

3.1.6 Relation to This Thesis

Mask-Predict is the intellectual ancestor of the entire thesis. It demonstrates empirically that confidence-guided iterative remasking works, but provides no explanation for *why*. The thesis supplies the missing theoretical foundation: within the MDM framework, the max-probability confidence signal $c^i = \max_j p_\theta^i$ is an approximation of the true information content $I^i(x) = H(x^i \mid x^{\ i})$; remasking positions with low c^i is equivalent to locally refining the Riemann approximation of the information profile integral; and confidence-guided remasking reduces E_{fact} under conditions that are now formally stated.

The thesis generalises Mask-Predict in three ways: (1) it embeds iterative remasking in the MDM probabilistic framework with a proper forward process and reverse kernel; (2) it considers entropy and margin as alternative confidence signals, not just max-probability; (3) it derives conditions under which each signal is sufficient for principled remasking. Citing Mask-Predict anchors the thesis in the broader NAR generation literature and frames the contribution as *formalising a successful heuristic*.

3.1.7 Study Questions

1. Mask-Predict re-masks the n_t *lowest-confidence* tokens per step. Why lowest and not highest? What would happen if you re-masked the highest-confidence tokens instead?
2. The number of re-masked tokens follows the linear schedule $n_t = n(T - t)/T$. Is there a principled justification for this schedule, or is it purely heuristic? How does it relate to the EB-Sampler's entropy budget?
3. Mask-Predict is a conditional model (source $\to$ target). For unconditional MDM generation (no source), what replaces the "source context"? Does the confidence signal change interpretation?
4. The confidence signal $c^i = \max_j p_\theta^j$ (max-probability) is Tier 3. Under what condition does it rank positions in the same order as the true $I^i(x)$? When does the ranking diverge?

4 Part II — Discrete Diffusion: Foundations and Scaling

4.1 D3PM (2021) — The Origin of Absorbing Discrete Diffusion

Full citation: Austin, Johnson, Ho, Tarlow & van den Berg (2021), "Structured Denoising Diffusion Models in Discrete State-Spaces." *NeurIPS 2021.* arXiv:2107.03006.

4.1.1 Summary

D3PM is the foundational paper that brings diffusion models to discrete data, including text, graphs, and images with discrete pixel values. Continuous diffusion models (DDPM, score-based models) rely on Gaussian noise, which has no meaningful analogue for categorical variables. D3PM solves this by defining a *family* of discrete forward Markov chains parametrised by a transition matrix Q_t, and deriving a variational lower bound (VLB) that can be optimised to train the reverse process. *(Markov chain background: Appendix A.4; ELBO/VLB derivation: Appendix A.3.)*

The key design choice is the transition matrix Q_t. D3PM explores three options: (1) **absorbing diffusion** — tokens eventually collapse to a single absorbing [MASK] state — which is the process used by all subsequent MDMs; (2) **uniform diffusion** — tokens uniformly transition to all other tokens — which corresponds to gradual information loss but no absorbing state; (3) **embedding-based diffusion** — transitions follow distances in a learned embedding space.

For text, D3PM finds that the absorbing process significantly outperforms the alternatives. The reason is simple: masking preserves the structure of unmasked tokens entirely (they are unchanged), while uniform diffusion corrupts all tokens at every step, making it harder for the model to use the partially observed context. This empirical finding is the justification for the design choices of MDLM, SEDD, LLaDA, and all subsequent MDMs.

4.1.2 Method Details

General forward process. For a token $x^i \in \{0, 1, \dots, V-1\}$, the forward Markov chain is:

$$q(z_t^i \mid z_{t-1}^i) = \mathrm{Cat}(z_t^i;\ \bar{Q}_t^{z_{t-1}^i})$$

where $\bar{Q}_t = Q_1 Q_2 \cdots Q_t$ is the cumulative transition matrix and the superscript z_{t-1}^i denotes the row indexed by the current state.

Absorbing transition matrix. With mask token m as the absorbing state:

$$[Q_t]_{ij} = \begin{cases} 1-\beta_t & \text{if } i = j \neq m \\ \beta_t & \text{if } j = m,\ i \neq m \\ 1 & \text{if } i = j = m \end{cases}$$

The cumulative marginal has the clean form:

$$q(z_t^i \mid x^i) = \bar{\alpha}_t \cdot e_{x^i} + (1-\bar{\alpha}_t) \cdot e_m, \quad \bar{\alpha}_t = \prod_{s=1}^{t} (1-\beta_s)$$

which is exactly the MDLM forward process with $\alpha_t = \bar{\alpha}_t$. *(Full derivation of the absorbing marginal and posterior: Appendix A.6.)*

Posterior (closed form for absorbing diffusion). Given z_t^i and the clean token x^i:

$$q(z_{t-1}^i \mid z_t^i, x^i) = \begin{cases} 1 & \text{if } z_t^i = x^i \text{ (token is already clean)} \\ \frac{\bar{\alpha}_{t-1}(1-\beta_t)}{\bar{\alpha}_t} \cdot e_{x^i} + \frac{(1-\bar{\alpha}_{t-1})\beta_t + (1-\bar{\alpha}_t)(1-\beta_t)}{1-\bar{\alpha}_t} \cdot e_m & \text{if } z_t^i = m \end{cases}$$

For the absorbing process this simplifies to: if $z_t^i \neq m$, then $z_{t-1}^i = z_t^i$ with certainty; if $z_t^i = m$, then $z_{t-1}^i = x^i$ with probability $\bar{\alpha}_{t-1}/(1 - \bar{\alpha}_t \cdot \mathbf{1}_{z_t^i \neq m})$ and $z_{t-1}^i = m$ otherwise. This is the key posterior used by all subsequent MDMs.

VLB training objective. D3PM minimises the variational lower bound *(ELBO derivation: Appendix A.3; KL divergence: Appendix A.2.3)*:

$$\mathcal{L}_{\text{VLB}} = \underbrace{\mathbb{E}[\text{KL}(q(z_T \mid x) \| p(z_T))]}_{\text{prior matching}} + \sum_{t=2}^{T} \mathbb{E}[\text{KL}(q(z_{t-1} \mid z_t, x) \| p_\theta(z_{t-1} \mid z_t))] + \mathbb{E}[-\log p_\theta(x \mid z_1)]$$

The x_0-parameterisation predicts $\tilde{p}_\theta(x \mid z_t)$ (the clean token distribution) and then uses the closed-form posterior to compute $p_\theta(z_{t-1} \mid z_t)$. An auxiliary loss $\lambda \cdot \mathbb{E}[\text{CE}(x, \tilde{p}_\theta(z_t))]$ directly supervises the clean-token prediction. MDLM later shows this auxiliary term dominates and the VLB can be simplified to this term alone.

4.1.3 Theoretical Contributions

D3PM proves that the VLB for discrete Markov chains decomposes into a sum of per-step KL terms, analogously to DDPM for continuous diffusion. It establishes that the posterior $q(z_{t-1} \mid z_t, x)$ has a tractable closed form for any transition matrix whose cumulative product $\bar{Q}_t$ can be computed in closed form (which is the case for absorbing and uniform matrices). The paper does not provide non-asymptotic sampling error bounds; the error analysis comes later with Lavenant & Zanella (2025).

4.1.4 Confidence Signal

D3PM uses ancestral sampling with no confidence signal. At each step, it samples $z_{t-1} \sim p_\theta(z_{t-1} \mid z_t)$ uniformly over all masked positions. The x_0-parameterisation (predicting $\tilde{p}_\theta(x \mid z_t)$) is the source of all subsequent per-token logit distributions that confidence signals are derived from.

4.1.5 Limitations

1. **Absorbing process chosen empirically.** D3PM shows absorbing > uniform > embedding-based on text, but provides no theoretical derivation of which Q_t is optimal. The Lavenant & Zanella framework later provides a partial answer: the absorbing process minimises E_{fact} for any given information profile.
2. **Slow sampling.** D3PM uses $T = 1000$ steps (borrowed from DDPM). MDLM reduces this to $T = 32$–1024 with no loss of quality.

3. **No remasking.** The ancestral sampler is strictly one-directional: a committed token stays committed.
4. **Sequence-level independence.** Positions are treated as conditionally independent given z_t in the forward process, which is the root of E_{fact}.
5. **Limited language modelling experiments.** D3PM evaluates on character-level text8, not word-level OpenWebText; MDLM later shows the framework scales to the harder word-level setting.

4.1.6 Relation to This Thesis

D3PM is the origin of the entire MDM framework used in the thesis. The absorbing transition matrix Q_t introduced here is the source of E_{fact}: positions are masked independently in the forward process, so the true reverse-step posterior has inter-token correlations that the factored model $p_\theta(z_{t-1} \mid z_t) = \prod_i p_\theta(z^i_{t-1} \mid z_t)$ cannot capture. The thesis's remasking kernel corrects this: by re-masking a high-entropy committed token, the model re-predicts it in a context that may have changed since the token was first committed, partially recovering the inter-token dependencies that the independence assumption discarded.

4.1.7 Study Questions

1. D3PM evaluates three transition matrices: absorbing, uniform, and embedding-based. Why does absorbing outperform uniform for text? What property of the absorbing process is valuable and what does uniform destroy?
2. The posterior $q(z^i_{t-1} \mid z^i_t, x^i)$ is tractable in closed form. Write it out for $z^i_t = m$ (masked) and $z^i_t = x^i$ (already committed). What is the probability that a masked token remains masked at the next step?
3. D3PM uses an auxiliary cross-entropy loss $\lambda \cdot \text{CE}(x, \tilde{p}_\theta(z_t))$. MDLM later shows this term dominates the VLB. Intuitively, why would directly predicting x from z_t be sufficient, without the VLB terms?
4. D3PM has no remasking. If you wanted to add a single remasking step after standard ancestral sampling, where in the algorithm would you insert it and what would determine which token to remask?

4.2 SEDD (2024) — Score Entropy and Predictor-Corrector Samplers

Full citation: Lou, Meng & Ermon (2024), "Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution." *ICML 2024.* arXiv:2310.16834.

4.2.1 Summary

SEDD introduces an alternative training objective for discrete diffusion models that is more theoretically principled than the VLB of D3PM. Instead of parameterising the reverse Markov kernel $p_\theta(z_{t-1} \mid z_t)$ directly, SEDD parameterises the *concrete score*:

$$s_\theta(z_t, j) \approx \frac{p_t(z_t^{(j)})}{p_t(z_t)}$$

where $z_t^{(j)}$ is the sequence z_t with one specific position changed to token j, and p_t is the marginal distribution at time t. This is the discrete analogue of the score function $\nabla_x \log p_t(x)$ in continuous diffusion. SEDD trains this score with a *score entropy* objective that is a proper scoring rule and provably consistent.

The key practical contribution is the **predictor-corrector (PC) sampler**: at each denoising step, SEDD first applies a standard predictor (reverse process) and then applies a Langevin-type corrector that uses the score to refine the current sequence. This corrector is the discrete diffusion analogue of the remasking step in ReMDM and is the principled precursor to all remasking approaches.

4.2.2 Method Details

Score entropy objective. The concrete score is trained by minimising:

$$\mathcal{L}(\theta) = \mathbb{E}_{t,x,z_t}\left[\sum_{j \neq z_t^i} w_t \cdot \left(s_\theta(z_t)_{z_t^i,j} - \log \frac{p_t(z_t^{(j)})}{p_t(z_t)} + h(s_\theta(z_t)_{z_t^i,j})\right)\right]$$

where $h(u) = e^u - u - 1 \geq 0$ is a convex function that, together with the log-ratio term, forms a proper scoring rule. At the optimal θ^*, $s_\theta(z_t)_{ij} = p_t(z_t^{(j)})/p_t(z_t)$ exactly.

Equivalence with MDM for absorbing noise. For the absorbing (masking) noise process, the score $s_\theta(z_t)_{m,j}$ for a masked position simplifies to $p_\theta(x^i = j \mid z_t)/p_t(z_t^i = j)$. Since the denominator $p_t(z_t^i = j)$ is a constant (determined by the marginal noise level), estimating the score is equivalent to estimating $p_\theta(x^i = j \mid z_t)$ — the same objective as MDLM's masked cross-entropy. This explains why MDLM achieves the same quality as absorbing SEDD despite its simpler objective.

Predictor-corrector sampler. At each step $t \to t-1$:

Predictor: $z' \leftarrow p_\theta(z_{t-1} \mid z_t)$ (standard reverse step, unmasks a fraction of tokens).

Corrector (K steps): Repeat K times:

```
for k = 1, ..., K:
    for each position i:
        # Compute transition rate using the concrete score
        rate_{z'^i → j} = s_ (z')_{z'^i, j} · R_t(z'^i, j)
        # Apply one step of the continuous-time Markov chain
        z'^i ← transition according to rate
```

where R_t is the noise process transition rate matrix (the time derivative of Q_t). The corrector steps can be understood as approximately sampling from $q_t(x \mid z_t)$ using MCMC with the learned score as the energy function.

Tweedie denoiser. SEDD derives the discrete analogue of Tweedie's formula, giving the minimum mean squared error (MMSE) estimate of x given z_t directly from the score. This MMSE estimate can be used as a deterministic initialisation before stochastic sampling.

4.2.3 Theoretical Contributions

SEDD provides two formal results:

1. **Consistency of the score entropy objective.** The minimiser θ^* of $\mathcal{L}(\theta)$ satisfies $s_{\theta^*}(z_t)_{ij} = p_t(z_t^{(j)})/p_t(z_t)$ for all z_t, i, j — i.e., the score is exactly recovered at optimality. This is the discrete analogue of Fisher consistency in score matching.
2. **PC sampler convergence.** Under mild conditions on the noise schedule, the PC sampler converges to the true data distribution as $T \to \infty$ and $K \to \infty$, with an explicit KL convergence rate. This is a stronger guarantee than D3PM's VLB bound (which only bounds training, not sampling quality) and predates Lavenant & Zanella (2025).

4.2.4 Confidence Signal

SEDD does not use a per-token confidence signal for *selective* correction. The corrector step applies the CTMC transition uniformly to all positions. The concrete score $s_\theta(z_t)_{z_t^i,j}$ encodes how strongly the model "wants" to change position i from its current value to j; positions with high total outgoing rate $\sum_j s_\theta(z_t)_{z_t^i,j}$ are high-entropy and natural candidates for correction. This is the Tier 1 signal (true posterior ratio) that the thesis's Tier 3 heuristics approximate.

4.2.5 Limitations

1. **Score entropy objective is harder to implement** than cross-entropy. For the absorbing process, the two objectives are equivalent, removing SEDD's theoretical advantage on the most practically important case.
2. **PC corrector is not selective.** The corrector applies to all positions, not just those where correction is most beneficial. This makes it less efficient than a confidence-guided corrector.
3. **Non-absorbing SEDD underperforms.** The uniform noise process gives worse text quality than absorbing noise, confirming that absorbing is the right inductive bias for language.
4. **No per-token confidence signal for unmasking order.** SEDD unmasks positions in the default schedule order, not in order of information content.

4.2.6 Relation to This Thesis

SEDD's PC sampler is the most direct precursor to the remasking approach of this thesis. The corrector step in SEDD — applying discrete CTMC transitions guided by the concrete score — is a principled correction mechanism that can in principle target high-entropy positions. The thesis extends this in two ways: (1) it derives an explicit E_{fact} reduction bound for selective correction (rather than the uniform CTMC correction of SEDD); (2) it shows that the entropy-based Tier 3 signals (which require only the forward-pass logits, not the full concrete score) are sufficient for achieving this reduction under a consistency assumption. The SEDD submodule (`external/sedd`) provides pretrained checkpoints on OpenWebText for baseline comparison.

4.2.7 Study Questions

1. For the absorbing noise process, the concrete score simplifies to $s_\theta(z_t)_{m,j} \propto p_\theta(x^i = j \mid z_t)$. This means SEDD's objective is equivalent to MDLM's cross-entropy for absorbing noise. Does this mean SEDD and MDLM learn the same model? What differences remain?
2. SEDD's corrector applies CTMC transitions to *all* positions uniformly. The thesis applies remasking selectively. Formally, what would a "selective SEDD corrector" look like? Would it still have SEDD's convergence guarantee?
3. SEDD proves the PC sampler converges as $T \to \infty$ and $K \to \infty$. Is there a finite-T, finite-K convergence rate? What does it depend on?

4. The concrete score $s_\theta(z_t)_{ij} = p_t(z_t^{(j)})/p_t(z_t)$ is a ratio of marginals. At what point in the generation process is this ratio most informative? At the beginning (many masked tokens) or the end (few masked tokens)?

4.3 MD4 and MDLM (2024) — The Practical MDM Framework

Full citations: Shi et al. (2024), "Simplified and Generalized Masked Diffusion for Discrete Data" (MD4); Sahoo et al. (2024), "Simple and Effective Masked Diffusion Language Models" (MDLM). *NeurIPS 2024.* arXiv:2406.07524.

4.3.1 Summary

MD4 and MDLM simultaneously simplify and scale the D3PM framework, arriving at what is now the standard MDM formulation. Both papers observe that the D3PM VLB simplifies dramatically for the absorbing noise process: the dominant training signal is a weighted sum of per-position cross-entropy losses on masked tokens, and the weighting can be chosen via importance sampling to minimise variance. MDLM additionally shows that a cosine noise schedule achieves approximately uniform signal-to-noise ratio per step, improving training stability.

MDLM-OWT (130M parameters, trained on OpenWebText) is the experimental backbone of the thesis. It is fast (single forward pass ~10ms on A100), publicly available (`kuleshov-group/mdlm`), and its architecture is the base that ReMDM, PRISM, and many other systems build on.

4.3.2 Method Details

Forward process. Identical to D3PM's absorbing process:

$$q(z_t \mid x) = \prod_i \mathrm{Cat}(z_t^i;\ \alpha_t \cdot e_{x^i} + (1-\alpha_t)\cdot e_m)$$

Noise schedules. MD4 uses linear $\alpha_t = 1 - t$. MDLM introduces the cosine schedule $\alpha_t = \cos^2(\pi t/2)$, which keeps α_t close to 1 for most of $[0,1]$ and drops sharply near $t = 1$, concentrating training signal at moderate noise levels where the model's predictions are neither trivially easy nor impossible.

Reverse kernel. From D3PM's posterior (Bayes' rule on the absorbing process):

$$p_\theta(z_s^i \mid z_t) = \frac{\alpha_s}{\alpha_t}\cdot \delta(z_s^i = z_t^i) + \left(1 - \frac{\alpha_s}{\alpha_t}\right)\cdot p_\theta(x^i \mid z_t)$$

For $z_t^i = m$ (masked): unmask with probability $1 - \alpha_s/\alpha_t$, re-masking otherwise. For $z_t^i \neq m$ (already committed): stay with probability 1. This is the "absorbing is absorbing" property: no remasking in the standard process.

MDLM training objective. The simplified ELBO is:

$$\mathcal{L}_{\mathrm{MDLM}} = \mathbb{E}_{t\sim p(t),\, x\sim\pi,\, z_t\sim q_t(\cdot|x)}\left[w_t \cdot \sum_{\{i:\, z_t^i = m\}} -\log p_\theta(x^i \mid z_t)\right]$$

where $w_t = \alpha_t'/(1-\alpha_t)$ is the importance weight derived from the noise schedule derivative. This is simply a weighted masked cross-entropy, identical to BERT pre-training but with a principled per-step weighting.

Sampling algorithm. Standard MDLM sampling with T steps:

```
z_T = [m, m, ..., m]
for t = T, T-1, ..., 1:
    # One forward pass: get clean token predictions for all masked positions
    p^i = p_ (x^i | z_t)    for all i with z_t^i = m
    # Sample: unmask each masked position with probability (1 -  _{t-1}/ _t)
    for i with z_t^i = m:
        if Uniform(0,1) < 1 -  _{t-1}/ _t:
            z_{t-1}^i ← sample from p^i        # commit
        else:
            z_{t-1}^i ← m                       # stay masked
return z_0
```

This is essentially: at each step, independently commit each masked position with probability proportional to the noise schedule derivative. The *order* of commitments is random and position-independent — there is no priority given to easy tokens.

4.3.3 Theoretical Contributions

MDLM proves that the simplified objective $\mathcal{L}_{\text{MDLM}}$ is a strict lower bound on the VLB $\mathcal{L}_{\text{D3PM}}$ and that the difference goes to zero as the model approaches the true score. This justifies replacing the VLB with the simpler objective. The cosine schedule is motivated by showing that $\text{SNR}(t) = \alpha_t/(1-\alpha_t)$ has approximately constant log-derivative under the cosine schedule, which is the MDM analogue of the continuous diffusion result that uniform SNR spacing minimises training variance. No non-asymptotic sampling error bounds are provided.

4.3.4 Confidence Signal

None at inference time. The per-position distribution $p_\theta(x^i \mid z_t)$ is available for free at every step and is the raw material for all confidence signals in the thesis: - Max-prob: $c^i = \max_j p_\theta(x^i = j \mid z_t)$ - Entropy: $c^i = -H(p_\theta(x^i \mid z_t)) = \sum_j p_\theta^j \log p_\theta^j$ - Margin: $c^i = p_\theta^{(1)} - p_\theta^{(2)}$ (gap between top-2)

4.3.5 Limitations

1. **Uniform random unmasking order.** Positions are committed independent of their uncertainty, violating the optimality condition derived by Lavenant & Zanella.
2. **No remasking.** The absorbing kernel sets $z_s^i = z_t^i$ with probability $\alpha_s/\alpha_t > 0$ only for *unmasked* positions; masking an already-committed token requires going forward in the noise process, which the reverse sampler does not do.
3. **Independence assumption.** The factored reverse kernel $\prod_i p_\theta(z_s^i \mid z_t)$ ignores the true joint posterior, inducing E_{fact}.
4. **No formal sampling error analysis.** Quality guarantees are purely empirical (perplexity on OWT), not formal.

4.3.6 Relation to This Thesis

MDLM-OWT is the primary experimental backbone. The forward process, reverse kernel, and training objective of MDLM are the setting in which the thesis's theoretical results are derived. The per-token logit distribution $p_\theta(x^i \mid z_t)$ is the source of all training-free confidence signals. The independence assumption of the reverse kernel is the root of E_{fact}, the quantity the thesis bounds.

4.3.7 Study Questions

1. The MDLM reverse kernel reads: "if $z_t^i = m$, unmask with probability $1 - \alpha_s/\alpha_t$". What is the probability when $s = t - 1$ and the cosine schedule is used? How does this change between early steps ($t \approx T$) and late steps ($t \approx 1$)?
2. MDLM commits tokens uniformly at random, ignoring position uncertainty. Using the Lavenant & Zanella bound, what is the E_{fact} penalty paid by committing a high-entropy token i (with $I^i \gg \bar{I}$) early?
3. The cosine schedule keeps α_t near 1 for most of $[0, 1]$ and drops sharply near $t = 1$. How does this affect how many tokens are unmasked per step at different stages? Is there a step count where most tokens are committed?
4. MDLM's training loss is weighted masked cross-entropy. If you had unlimited data and a perfectly trained model ($p_\theta = q$), would E_{fact} be zero? Why or why not?

4.4 LLaDA (2025) — Scaling Masked Diffusion to 8B Parameters

Full citation: Nie et al. (2025), "LLaDA: Large Language Diffusion with mAsking." arXiv:2502.09992. HuggingFace: `GSAI-ML/LLaDA-8B-Instruct`.

4.4.1 Summary

LLaDA answers the question: can masked diffusion language models compete with frontier autoregressive LLMs at scale? Training an 8B-parameter bidirectional transformer (LLaDA-8B) on 2.3 trillion tokens with the MDLM objective, the paper shows competitive or superior performance to LLaMA 3 8B on MMLU, GSM8K, HumanEval, and multi-turn dialogue — the first masked diffusion model to achieve this at any scale. The instruction-tuned variant (LLaDA-8B-Instruct) is the backbone for PRISM's code generation experiments and is one of the two experimental backbones (alongside MDLM-OWT) of the thesis.

The key implication for the thesis: LLaDA demonstrates that MDMs are not a niche academic curiosity but a genuine frontier model architecture, and that improving their sampling quality has practical significance. Crucially, LLaDA uses a naive uniform remasking schedule — no confidence signal — leaving substantial quality on the table at inference time.

4.4.2 Method Details

Architecture. LLaDA-8B is a standard transformer with: - Bidirectional (non-causal) self-attention — unlike causal LLMs, every token can attend to every other token, which is natural for the masked prediction objective. - Pre-LayerNorm and RoPE positional encodings (standard modern LLM recipe). - Vocabulary: 128K tokens (LLaMA 3 tokeniser for comparability). - 32 layers, 32 heads, hidden dimension 4096 (same as LLaMA 3 8B).

Pre-training. Identical to MDLM: the loss is the importance-weighted masked cross-entropy on the 2.3T token corpus. The noise schedule follows MDLM's cosine schedule.

Instruction tuning (LLaDA-8B-Instruct). Given a prompt x and response y, the model is trained with the response tokens masked and the prompt tokens frozen:

$$\mathcal{L}_{\text{SFT}} = \mathbb{E}_{t,y_t}\left[w_t \sum_{\{i:\, y_t^i=m\}} -\log p_\theta(y^i \mid x, y_t)\right]$$

This is the first demonstration of SFT for MDMs working at 8B scale, and the recipe is clean: just mask the response tokens and train the MDLM objective conditioned on the unmasked prompt.

Sampling. LLaDA uses a stochastic remasking schedule during generation: at each step t, all committed (non-masked) tokens are independently re-masked with probability $\rho(t)$, where $\rho(t)$ is a fixed schedule not conditioned on token confidence. The motivation is that remasking allows the model to correct early commitments in later steps when more context is available — but the schedule is heuristic, not principled.

4.4.3 Theoretical Contributions

LLaDA's contribution is empirical: it demonstrates scale, not theory. The paper provides scaling analysis showing that LLaDA obeys similar scaling laws to GPT-style models (loss $\propto$ compute$^{-\beta}$), which validates MDMs as a compute-competitive alternative to autoregressive LLMs. No theoretical results on sampling quality.

4.4.4 Confidence Signal

None. LLaDA's remasking schedule is based only on the current time step t (a fixed function), not on per-token confidence. This is the thesis's key target: replacing LLaDA's uniform schedule with a confidence-guided one should improve quality without any fine-tuning.

4.4.5 Limitations

1. **Naive remasking.** Uniform schedule ignores which tokens are uncertain. This is the primary weakness the thesis addresses.
2. **Bidirectional attention only.** Cannot be used for streaming or left-to-right generation, limiting some deployment contexts.
3. **PRISM adapter not public.** The PRISM quality head trained on LLaDA-8B is unavailable publicly (as of early 2026), limiting comparison to the PRISM upper bound.
4. **No formal connection to information profile.** The remasking schedule is not derived from $I(x)$ or any principled criterion.

4.4.6 Relation to This Thesis

LLaDA-8B-Instruct is the large-scale experimental backbone. The main experiment on LLaDA is: does replacing LLaDA's uniform remasking with the thesis's confidence-guided strategies (entropy threshold, top-k low-confidence) improve generation quality on standard benchmarks (LAMBADA, HellaSwag, MBPP)? If yes at 8B scale, the thesis's contribution generalises beyond the small MDLM-OWT setting. The uniform remasking of LLaDA is the weakest baseline the thesis should beat by a significant margin.

4.4.7 Study Questions

1. LLaDA's remasking schedule re-masks committed tokens with probability $\rho(t)$ at each step, independent of token confidence. The thesis uses confidence-guided remasking. LLaDA is 8B parameters vs MDLM's 130M. Do you expect the benefit of confidence-guided remasking to be *larger* or *smaller* at 8B scale? Why?
2. LLaDA uses bidirectional (non-causal) attention. What does this mean for how the model uses context when predicting a masked token? How does it differ from GPT-style causal LLMs?
3. LLaDA's instruction tuning freezes prompt tokens and only masks response tokens. Is the information profile $I^i(x)$ for response tokens affected by whether the prompt is long or short? What would you expect for very long prompts?
4. LLaDA achieves competitive performance with LLaMA 3 8B on MMLU and GSM8K. Why might a masked diffusion model be competitive on these tasks despite generating text in a non-autoregressive manner?

5 Part III — A Unified Theoretical Lens: Discrete Flow Matching

5.1 Discrete Flow Matching (2024) — DFM as a Unifying Framework

Full citation: Gat, Remez, Shaul, Kreuk, Chen, Synnaeve, Adi & Lipman (2024), "Discrete Flow Matching." *NeurIPS 2024.* arXiv:2407.15595.

5.1.1 Summary

Discrete Flow Matching (DFM) extends continuous flow matching (Lipman et al., 2022) to categorical data, providing a unified theoretical framework that subsumes both MDMs (D3PM, MDLM) and SEDD as special cases. The key insight is that generative modelling of discrete data can be understood as learning a probability flow from a simple prior (e.g., uniform or fully masked) to the data distribution, defined by a *velocity field* on the space of probability simplices.

For the thesis, DFM provides two valuable things. First, it shows that the masked cross-entropy training objective of MDLM is exactly the flow matching objective under the absorbing probability path — this unifies the training objectives across the literature and removes the need to reason about VLBs or score entropy separately. Second, DFM gives the cleanest language for corrector steps: a corrector is a flow that leaves the marginal distribution p_t invariant while reducing entropy along individual trajectories, which corresponds precisely to re-masking high-entropy tokens and re-predicting them.

5.1.2 Method Details

Probability paths. DFM defines a parameterised family of probability paths $p_t(z)$ interpolating between a prior p_0 and the data distribution p_1. For the absorbing path (the MDM case):

$$p_t(z) = \sum_{x \sim p_{\text{data}}} \text{Cat}(z;\ \alpha_t \cdot e_x + (1 - \alpha_t) \cdot e_m)$$

This is identical to the MDLM forward-process marginal.

Discrete velocity field. The generative model is a continuous-time Markov chain (CTMC) over sequences, specified by a rate matrix. For the absorbing path, the velocity field (rate of probability flow from mask token m to token j at position i) is:

$$u_t(z^i = m \to j;\ x) = \frac{\dot{\alpha}_t}{1 - \alpha_t} \cdot \mathbf{1}[j = x^i]$$

where $\dot{\alpha}_t = d\alpha_t/dt < 0$. The marginalised velocity (averaging over x given z_t) is:

$$u_t(z^i = m \to j) = \frac{-\dot{\alpha}_t}{1 - \alpha_t} \cdot p_\theta(x^i = j \mid z_t)$$

This is the quantity the model p_θ learns to estimate; minimising the discrete flow matching objective is equivalent to minimising the masked cross-entropy of MDLM (up to a time-weighting factor).

Training objective. The discrete flow matching loss for the absorbing path:

$$\mathcal{L}_{\text{DFM}} = \mathbb{E}_{t,x,z_t}\left[\frac{-\dot{\alpha}_t}{1-\alpha_t}\sum_{\{i:\, z_t^i=m\}} -\log p_\theta(x^i \mid z_t)\right]$$

This is exactly the MDLM objective with time weight $w_t = -\dot{\alpha}_t/(1-\alpha_t)$.

Corrector steps in DFM. A corrector for a discrete flow model is any CTMC that: (1) leaves p_t stationary; and (2) reduces entropy along individual sequences. The canonical choice is a Gibbs sampler with stationary distribution p_t: pick a position i, mask it ($z^i \leftarrow m$), and re-sample from $p_\theta(x^i \mid z_t)$. This is exactly the remasking operation of ReMDM.

The DFM perspective makes clear *why* this works: the Gibbs corrector moves the current sequence z closer to a high-probability sequence under p_t while preserving the marginal distribution. The thesis's contribution in DFM language is: a *confidence-guided* Gibbs corrector that selectively applies this operation to high-entropy positions reduces the KL between the current trajectory distribution and p_t more efficiently than a uniform Gibbs corrector.

5.1.3 Theoretical Contributions

1. **Unification.** DFM proves that MDMs and SEDD are both special cases of the discrete flow matching framework with different choices of probability path.
2. **Objective equivalence.** For the absorbing path, the DFM objective is identical to the MDLM cross-entropy (up to weighting), justifying MDLM's simple training.
3. **Optimal probability path.** DFM derives the path that minimises the continuous-time KL between the generative trajectory and the data-generating process, providing a theoretical basis for choosing the noise schedule.
4. **Corrector theory.** DFM gives the cleanest theoretical framework for corrector steps: they are Gibbs samplers for p_t, and their mixing rate is the spectral gap of the Gibbs chain.

5.1.4 Confidence Signal

DFM does not introduce a new confidence signal. The velocity field $u_t(z^i = m \to j)$ is proportional to $p_\theta(x^i = j \mid z_t)$, so the entropy of the velocity field $H(u_t(z^i = m \to \cdot))$ equals the entropy of the token prediction distribution $H(p_\theta(x^i \mid z_t))$ — the Tier 3 entropy signal of the thesis.

5.1.5 Limitations

1. **Continuous-time formulation** is cleaner theoretically but harder to implement than the discrete-step MDLM sampler; in practice, the Euler discretisation of the DFM ODE reduces to MDLM sampling.
2. **Corrector theory is not quantitative.** DFM shows that correctors work but does not derive an explicit bound on the KL reduction per corrector step, and does not connect to E_{fact}.
3. **Limited text experiments.** DFM evaluates primarily on protein sequences and audio tokens; text results are limited compared to MDLM-OWT.
4. **The absorbing path is not proven optimal.** It is one choice among infinitely many; the optimal path for text generation is an open question.

5.1.6 Relation to This Thesis

DFM provides the cleanest theoretical language for Chapter 3 of the thesis. The thesis's main theorem — confidence-guided remasking reduces E_{fact} — can be stated in DFM language as: a *confidence-guided Gibbs corrector* reduces the discretisation error of the DFM Euler sampler by targeting positions where the velocity field has the highest entropy. Including DFM in the literature review positions the thesis at the intersection of diffusion theory and flow matching, broadening the potential readership and citation impact.

5.1.7 Study Questions

1. DFM shows the training objective is $\mathcal{L}_{\text{DFM}} \propto \frac{-\dot{\alpha}_t}{1-\alpha_t} \sum_i -\log p_\theta(x^i \mid z_t)$. Compare this to MDLM's $w_t = \alpha_t'/(1-\alpha_t)$. Are these the same time-weighting? What does the DFM perspective add beyond MDLM?
2. A DFM corrector is any CTMC that leaves p_t stationary. Is a corrector that *only* re-masks tokens (but never unmasked them) a valid corrector in DFM? What condition would it need to satisfy?
3. The "optimal probability path" in DFM minimises the continuous-time KL. The absorbing path is one option. Give an intuition for what a "better" path might look like and why it might not be absorbing.
4. The mixing rate of the Gibbs corrector is the spectral gap λ. What determines λ for a text sequence? How does sequence length affect it? How does dependency structure (e.g., code vs. random text) affect it?

6 Part IV — The Error-Bound Framework: Why Algorithms Fail and How to Fix Them

6.1 Lavenant & Zanella (2025) — The Foundation: Decomposing Sampling Error

Full citation: Lavenant & Zanella (2025), "Sampling Error Analysis for Masked Diffusion Models."

6.1.1 Summary

This paper provides the first rigorous non-asymptotic analysis of the sampling error of MDM algorithms. Its central contribution is a KL divergence bound that decomposes the total sampling error into two independent terms: (1) a *learning error* E_{learn} due to the model p_θ not perfectly matching the true score, and (2) a *factorization error* E_{fact} due to the sampling algorithm committing multiple tokens simultaneously and treating them as independent. The factorization error is characterised as a Riemann approximation error of the sequence's *information profile*, and the paper derives the optimal unmasking schedule that minimises it.

The remasking direction — what happens when committed tokens are re-masked and re-predicted — is explicitly identified as an open problem and the central gap this thesis fills. This paper is supervised by Prof. Zanella, the thesis supervisor.

6.1.2 Method Details

Setup. Let π be the true data distribution, p_θ the trained MDM, and p_{alg} the distribution over sequences produced by running the MDM sampler for T steps. The main bound is:

$$\mathrm{KL}(\pi \| p_{\text{alg}}) \leq E_{\text{learn}} + E_{\text{fact}}$$

Learning error. E_{learn} is the cumulative KL between the true denoising posterior and the model's approximation, summed over all steps:

$$E_{\text{learn}} = \sum_{t=1}^{T} \mathbb{E}_{z_t \sim q_t} \left[\mathrm{KL}\Big(q(x \mid z_t) \,\Big\|\, p_\theta(x \mid z_t) \Big) \right]$$

This goes to zero as $p_\theta \to q$ (perfect training) and is independent of the sampling algorithm's choices (which tokens to unmask, in what order). It can be bounded by the training loss.

Factorization error. E_{fact} arises from the independence assumption in the reverse kernel:

$$p_\theta(z_{t-1} \mid z_t) = \prod_i p_\theta(z_{t-1}^i \mid z_t)$$

The true posterior is not factored: simultaneously unmasking k positions from the same masked sequence introduces correlations that the product-of-marginals approximation cannot capture. The paper shows:

$$E_{\text{fact}}(t) \approx (\Delta\alpha_t)^2 \cdot \sum_i \mathrm{Var}_{x \sim q(x|z_t)}[H(x^i \mid x^{\ i})]$$

where $\Delta\alpha_t = \alpha_{t-1} - \alpha_t$ is the step size in the noise schedule.

Information profile. The key object is the per-position conditional entropy *(entropy and conditional entropy defined in Appendix A.2.1–A.2.2)*:

$$I^i(x) = H(x^i \mid x^{\ i}) \quad \text{(conditional entropy of position } i \text{ given all others)}$$

and the total information $I(x) = \sum_i I^i(x)$. The sequence $\{I^i(x)\}_{i=1}^{L}$ is the *information profile* of x; it measures how hard each position is to predict from its context.

E_{fact} as a Riemann error. Summing over steps and taking the continuum limit, the total factorization error is a Riemann approximation error of $\int_0^1 I(x)\,dt$:

$$E_{\text{fact}} \leq C \cdot \sum_{t=1}^{T} (\Delta\alpha_t)^2 \cdot \Sigma^2 \approx C \cdot \frac{\Sigma^2}{T}$$

where $\Sigma^2 = \mathrm{Var}_i[I^i(x)]$ is the variance of the information profile across positions. *(Derivation and intuition for this bound: Appendix A.7.)* This is minimised when: (1) T is large (many steps), and (2) Σ^2 is small (information is evenly distributed across positions, so no step is much harder than another).

Optimal unmasking schedule. To minimise E_{fact}: - Choose T as large as the compute budget allows. - Unmask positions in **increasing order of** $I^i(x)$ — easy tokens first, hard tokens last — so that at each step the information released is approximately equal: $\Delta I_t = I(x)/T$ for all t. - In practice, $I^i(x)$ is approximated by $H(p_\theta(x^i \mid z_t))$ (the model's predicted entropy), which is the EB-Sampler algorithm.

Main theorem (informal). For any unmasking schedule with T steps:

$$E_{\text{fact}} \leq C \cdot \left(\frac{I(x)}{T}\right)^2$$

with equality for the uniform step size schedule. The optimal (non-uniform) schedule achieves:

$$E^*_{\text{fact}} \leq C \cdot \frac{I(x)^2}{T^2} \cdot \left(1 + \frac{\Sigma^2}{\bar{I}^2}\right)^{-1}$$

where $\bar{I} = I(x)/L$ is the mean information per position.

6.1.3 Theoretical Contributions

1. **First non-asymptotic KL bound for MDM sampling** — establishes that the gap between π and p_{alg} can be controlled.
2. **Decomposition into $E_{\text{learn}} + E_{\text{fact}}$** — cleanly separates the effect of model quality from algorithm design.

3. **Identification of $E_{\textbf{fact}}$ as a Riemann error** of $I(x)$ — connects the sampling algorithm to information theory.
4. **Optimal unmasking schedule** — first principled derivation of which tokens to unmask in what order.
5. $O(k^2)$ **penalty for simultaneous unmasking** — shows that committing k tokens at once is $k^2\times$ worse than committing them sequentially.

6.1.4 Confidence Signal

None (this is a theoretical analysis paper, not an algorithm paper). The analysis implies that the optimal confidence signal for unmasking order is $I^i(x) = H(x^i \mid x^{\ i})$, and that $H(p_\theta(x^i \mid z_t))$ is its practical approximation.

6.1.5 Limitations

1. **Remasking not covered.** This is the explicit open problem that the thesis fills. The framework analyses only the unmasking direction.
2. $E_{\textbf{learn}}$ **treated as a black box.** The bound on E_{learn} is in terms of the model's training loss, which is not further analysed.
3. **Assumes factored posterior.** The analysis uses the product-of-marginals approximation as a baseline; the bound could be tighter with a more careful approximation.
4. **Constant C not explicitly computed.** The bound is tight up to an unspecified constant depending on sequence length and vocabulary size.

6.1.6 Relation to This Thesis

This paper *is* the thesis's theoretical foundation and is supervised by the thesis advisor. The thesis's Chapter 3 extends the Lavenant & Zanella framework in one direction: it adds remasking transitions to the reverse process and derives:

1. A new KL decomposition that includes a *remasking error term* E_{remask}.
2. Conditions under which $E_{\text{remask}} < 0$ — i.e., remasking strictly reduces the total bound.
3. A characterisation of the optimal remasking schedule in terms of the confidence signal and the information profile.

The central thesis theorem is the remasking analogue of Lavenant & Zanella's optimal unmasking result.

6.1.7 Study Questions

1. The factorization error is $E_{\text{fact}} \leq C \cdot \Sigma^2/T$ for the uniform schedule. What happens to E_{fact} as $T \to \infty$? Does this mean infinite steps gives perfect generation? What limits quality as $T \to \infty$?
2. The bound says $E_{\text{fact}} \propto (\Delta\alpha_t)^2$. Why quadratic, not linear? Intuitively, why does committing k tokens at once cost k^2 rather than k?
3. The optimal unmasking schedule is "easy tokens first" (ascending I^i). Why easy first and not hard first? Give an intuitive argument — what goes wrong if you commit the hardest token (highest I^i) first?
4. E_{learn} is the error due to $p_\theta \neq q$. Is E_{learn} reduced by remasking, or only E_{fact}? What would a "remasking that reduces E_{learn}" look like — is it even possible without retraining?

6.2 EB-Sampler (2025) — Optimal Unmasking via Entropy Budgets

Full citation: Ben-Hamu et al. (2025), "EB-Sampler: Entropy-Bounded Sampling for Masked Diffusion Models."

6.2.1 Summary

The EB-Sampler is the algorithmic companion to the Lavenant & Zanella framework: it derives the concrete algorithm that minimises E_{fact} and achieves the optimal unmasking schedule. The key idea is to control the *amount of information* (entropy) unmasked at each step by adaptively choosing how many and which tokens to unmask, rather than committing to a fixed number per step. By unmasking tokens in ascending order of predicted entropy and stopping when the entropy budget ε is consumed, the EB-Sampler ensures each step releases approximately equal information — the condition for E_{fact} minimisation.

The empirical result is striking: the EB-Sampler achieves 2–3× speedup over standard fixed-step MDLM at the same quality, or significantly better quality at the same step count. For the thesis, the EB-Sampler is the "gold standard" for the unmasking direction — the thesis's task is to derive an equally principled standard for the *remasking* direction.

6.2.2 Method Details

Information profile approximation. The true per-position information $I^i(x) = H(x^i \mid x^{\,i})$ requires knowing x (which is unavailable at generation time). The EB-Sampler approximates it by:

$$\hat{I}^i(z_t) = H(p_\theta(x^i \mid z_t)) = -\sum_j p_\theta(x^i = j \mid z_t) \log p_\theta(x^i = j \mid z_t)$$

This is the entropy of the model's predicted token distribution at position i. High $\hat{I}^i$ means the model is uncertain; low $\hat{I}^i$ means the model is confident. At optimality ($p_\theta = q$), $\hat{I}^i(z_t) \to I^i(x)$ in expectation.

Entropy budget algorithm. At each step t:

```
# Compute predicted entropy for all currently masked positions
Î^i ← H(p_ (x^i | z_t))   for all i with z_t^i = m

# Sort by ascending entropy (easy positions first)
sorted_positions ← argsort(Î, ascending=True)

# Greedily unmask until entropy budget  is consumed
budget ←
unmask_set ← {}
for i in sorted_positions:
    if budget >= Î^i:
        unmask_set.add(i)
        budget -= Î^i
    else:
```

```
        break

# Commit the selected positions
for i in unmask_set:
    z_{t-1}^i ← sample from p_ (x^i | z_t)
    z_{t-1}^j ← z_t^j   for j   unmask_set
```

The total number of steps is adaptive: if the sequence has total predicted information $\hat{I}(z_t) = \sum_i \hat{I}^i$, then roughly $T \approx \hat{I}/\varepsilon$ steps are needed to unmask everything.

Entropy budget and E_{fact}. The paper proves (Theorem 2):

$$E_{\text{fact}}(\text{EB}) \leq \varepsilon \cdot \max_i \hat{I}^i$$

Compare to the uniform fixed-step sampler with T steps:

$$E_{\text{fact}}(\text{uniform}) \leq \frac{I(x)}{T} \cdot \max_i I^i(x)$$

Both bounds have the same structure: (entropy per step) $\times$ (max per-position entropy). The EB-Sampler automatically sets the step size to be ε everywhere, which is exactly the Riemann partition that minimises E_{fact} for a given total information budget.

Optimality. The paper proves (Theorem 3) that among all deterministic unmasking schedules (strategies that decide which tokens to unmask at each step based on the current state), the EB-Sampler achieves the minimum E_{fact} for a given total number of forward passes. No deterministic schedule can do better.

6.2.3 Theoretical Contributions

1. **Derives the EB-Sampler as the optimal unmasking algorithm** within the Lavenant & Zanella framework.
2. **Proves the E_{fact} bound** for the EB-Sampler explicitly.
3. **Proves optimality** among deterministic unmasking strategies.
4. **Connects entropy budgets to information profiles** — the first practical algorithm directly motivated by $I(x)$.

6.2.4 Confidence Signal

$\hat{I}^i = H(p_\theta(x^i \mid z_t))$ — the entropy of the predicted token distribution. This is a Tier 3 (heuristic, training-free) signal that is free to compute from the forward-pass logits. The EB-Sampler does not need a separate confidence model; the entropy is always available.

Why entropy and not max-prob or margin? Entropy is the most natural approximation of $I^i(x) = H(x^i \mid x^{\ i})$ because both measure "how uncertain is the distribution over x^i?". Max-probability and margin are monotonically related to entropy for unimodal distributions but diverge for multimodal ones (e.g., when the model is split between two plausible tokens).

6.2.5 Limitations

1. **Unmasking only.** The optimality result covers only the forward (unmasking) direction. Remasking is not considered, and there is no bound on what happens after errors are made.
2. $\widehat{I}^i$ **is a noisy proxy.** At early generation steps, the model's entropy estimate may poorly reflect the true $I^i(x)$ because the context is highly uncertain.
3. **Variable step count.** The adaptive number of steps makes it harder to compare against fixed-budget baselines on wall-clock time.
4. **No remasking.** Errors committed in early steps (when $\widehat{I}^i$ is poorly estimated) propagate to the end of generation.

6.2.6 Relation to This Thesis

The EB-Sampler and the thesis are direct complements:

$$\underbrace{\text{EB-Sampler}}_{\substack{\text{optimal unmasking}\\ \text{schedule}}} \quad \longleftrightarrow \quad \underbrace{\text{Thesis}}_{\substack{\text{optimal remasking}\\ \text{schedule}}}$$

The EB-Sampler minimises E_{fact} for the unmasking direction. The thesis minimises the additional E_{fact} reduction achievable by the remasking direction. The thesis's main theorem should state: given that the EB-Sampler has already produced a candidate sequence with some committed tokens, the remasking strategy that further reduces E_{fact} most efficiently is: remask the positions with the highest predicted entropy $\widehat{I}^i$, using threshold τ chosen to match the remaining entropy budget.

6.2.7 Study Questions

1. The EB-Sampler sets the entropy budget ε as the key hyperparameter. How should ε be chosen in practice? What is the relationship between ε and the total number of steps T?
2. The EB-Sampler bound is $E_{\text{fact}}(\text{EB}) \leq \varepsilon \cdot \max_i \widehat{I}^i$. The uniform bound is $E_{\text{fact}}(\text{uniform}) \leq (I(x)/T) \cdot \max_i I^i$. When does the EB-Sampler give a tighter bound? Is there a case where uniform is better?
3. The EB-Sampler uses $\widehat{I}^i(z_t) = H(p_\theta(x^i \mid z_t))$ which is an estimate of $I^i(x)$ conditioned on the *current noisy* z_t. Early in generation (high masking rate), how good is this estimate? Does $\widehat{I}^i$ converge to $I^i(x)$ as more tokens are committed?
4. The EB-Sampler commits tokens in ascending entropy order. After applying remasking (remasking some of the committed tokens), the ordering of the remaining masked tokens may change. Should the EB-Sampler re-sort all masked tokens after each remasking step? What is the computational cost of doing so?

7 Part V — Principled Remasking: From Heuristic to Learned to Optimal

---

7.1 ReMDM (2025) — The First Principled Remasking Posterior

Full citation: Wang, Schiff, Sahoo & Kuleshov (2025), "Remasking Discrete Diffusion Models with Inference-Time Scaling." arXiv:2503.00307. GitHub: `kuleshov-group/remdm`.

7.1.1 Summary

ReMDM is the paper that formally introduces remasking as an inference-time scaling mechanism for MDMs. It derives a principled *remasking posterior* σ_t — the probability of re-masking a committed token at time t — by analogy with the unmasking posterior of the standard reverse process. The key theoretical result is that adding correction passes (remask + re-predict) monotonically improves generation quality: more correction steps cannot hurt. The method requires no retraining and is evaluated on MDLM-OWT, with the ReMDM strategies (`remdm-conf`, `remdm-loop`, etc.) available in the submodule at `external/remdm`.

7.1.2 Method Details

The remasking posterior. The standard MDM reverse kernel specifies the probability of unmasking a masked token. ReMDM derives the complementary: the probability of *re-masking* a committed token at time t. Starting from first principles (what is the marginal probability that a clean token x^i is masked at time t, given that it was committed at some earlier time?), ReMDM derives:

$$\sigma_t^i = P(z_t^i = m \mid z_0^i = x^i) = 1 - \alpha_t$$

where α_t is the noise schedule evaluated at the current time. At high t (early in generation), $1 - \alpha_t$ is close to 1, so remasking is aggressive; at low t (near completion), $1 - \alpha_t$ is close to 0, so remasking is rare.

Inference algorithm with corrections:

```
z_T = [m, m, ..., m]
for t = T, T-1, ..., 1:
    # Standard predictor step (unmask)
    z_{t-1} ← sample from p_ (z_{t-1} | z_t)

    # R corrector steps (remask + re-predict)
    for r = 1, ..., R:
        # Re-mask each committed token independently
        for i with z_{t-1}^i   m:
            z'^i ← m   with probability  _t^i = 1 -  _t
            z'^i ← z_{t-1}^i   otherwise
        # Re-predict all masked positions
        z_{t-1} ← sample from p_ (z_{t-1} | z')
```

```
return z_0
```

Total forward passes: $T \times (R+1)$ — one per step plus R correction passes.

ReMDM strategies. The paper introduces four strategies for computing the remasking probability, going beyond the uniform σ_t:

- **`remdm-conf`** (confidence-weighted): remask token i with probability $\sigma_t^i \cdot (1 - c^i)$ where c^i is the model's confidence at that position. Low-confidence tokens are remasked more aggressively.
- **`remdm-cap`**: remask a fixed number of the lowest-confidence committed tokens per step.
- **`remdm-rescale`**: rescale the remasking probability to maintain a fixed expected number of remasked tokens.
- **`remdm-loop`** (time-windowed): only apply corrections when $t \in [t_{\text{off}}, t_{\text{on}}]$, concentrating corrections in the mid-generation window where they are most useful.

7.1.3 Theoretical Contributions

Monotone improvement theorem. Let $p^{(R)}_{\text{ReMDM}}$ be the distribution over sequences produced by ReMDM with R correction passes per step. Under the assumption that p_θ exactly matches the true score ($E_{\text{learn}} = 0$), ReMDM proves:

$$\mathrm{KL}(\pi \| p^{(R+1)}_{\text{ReMDM}}) \leq \mathrm{KL}(\pi \| p^{(R)}_{\text{ReMDM}}) \quad \forall R \geq 0$$

More correction steps always reduce the KL, monotonically. The proof uses the data-processing inequality applied to the Markov chain induced by the correction pass: the corrector can only reduce, not increase, the KL to the true distribution.

Inference-time scaling. As $R \to \infty$, $\mathrm{KL}(\pi \| p^{(R)}_{\text{ReMDM}}) \to 0$ for any $T \geq 1$. This is the "inference-time scaling" result: compute (number of forward passes) is directly tradeable against quality.

Gap from the thesis. ReMDM proves that corrections help but does not characterise *how much* they help per correction step, which corrections are most efficient (uniform vs. confidence-guided), or how the improvement relates to E_{fact} or the information profile. These are the thesis's contributions.

7.1.4 Confidence Signal

Baseline: uniform remasking with probability $\sigma_t = 1 - \alpha_t$ per committed token, independent of confidence. This is theoretically grounded (the monotone improvement theorem applies) but wasteful.

`remdm-conf`: remask with probability $\sigma_t \cdot (1 - c^i)$ where $c^i = H(p_\theta(x^i \mid z_t))$ (entropy). Empirically superior but no formal guarantee beyond monotone improvement.

7.1.5 Limitations

1. **No connection to E_{fact} or information profile.** The monotone improvement theorem is proven, but the *rate* of improvement per correction step is not characterised.
2. **Uniform baseline is wasteful.** Re-masking confident committed tokens with high probability wastes compute on positions that are unlikely to change.
3. **Compute cost is $O(T \times R)$.** For large R, this becomes prohibitive at 9B scale.
4. **Evaluation limited to language modelling.** LAMBADA and generative perplexity are the primary metrics; no instruction-following or reasoning benchmarks.

7.1.6 Relation to This Thesis

ReMDM is the closest existing work to the thesis. The σ_t posterior is a special case of the thesis's remasking kernel with $\tau = 0$ (always remask with probability $1 - \alpha_t$). The thesis: (1) derives an explicit E_{fact} reduction bound for the confidence-guided variant (showing *why* it is better, not just that it is); (2) derives the optimal threshold τ as a function of the information profile and entropy budget; (3) provides experiments on a broader set of models and benchmarks.

7.1.7 Study Questions

1. ReMDM's remasking probability is $\sigma_t^i = 1 - \alpha_t$. At step $t = T/2$ (halfway through generation), what is σ_t? Does remasking get more or less aggressive as generation progresses toward completion?
2. The monotone improvement theorem requires $E_{\text{learn}} = 0$ (perfect model). In practice, $p_\theta \neq q$. Does remasking still help when $E_{\text{learn}} > 0$? Could remasking ever *hurt* quality when the model is imperfect?
3. ReMDM's `remdm-conf` remasks with probability $\sigma_t(1 - c^i)$. Compare this to the thesis's threshold strategy (remask if $\hat{I}^i > \tau$). Are these equivalent? Which one commits to a hard decision and which uses a soft weighting?
4. Our experiments show remdm-conf MAUVE collapses at T=1000 while remdm-loop improves. Using the ReMDM framework, what is the remasking probability σ_t at T=1000 near the end of generation ($t \to 0$)? Is this consistent with the collapse hypothesis?

7.2 Informed Correctors (2024/2025) — MCMC Theory for Discrete Diffusion Correction

Full citation: Zhao et al. (2024/2025), "Informed Correctors for Discrete Diffusion Models."

7.2.1 Summary

Informed Correctors applies Markov Chain Monte Carlo (MCMC) theory to the correction problem for discrete diffusion models. The approach is Gibbs-based: at each correction step, one position is selected, re-masked, and re-predicted conditioned on all other positions. The *informed* variant selects positions in order of decreasing *surprise* (negative log-probability), so that the most uncertain tokens are corrected first.

The paper provides the strongest theoretical correction guarantee in the literature: an exponential mixing time bound showing that the KL between the corrected distribution and the target decays geometrically in the number of correction steps. It also proves that the informed (confidence-guided) corrector achieves a larger spectral gap than the uniform corrector — i.e., confidence-guided correction is strictly more efficient.

The key limitation for the thesis is that the exact informed corrector requires a *hollow transformer* — a modified attention mechanism that computes all per-position conditionals $q(z^d \mid z^{\ d})$ in a single forward pass. This requires architectural changes and retraining, making it incompatible with standard pretrained MDMs.

7.2.2 Method Details

Gibbs corrector. At each correction step, select a position d uniformly at random and resample:

$$z^d_{\text{new}} \sim q_t(z^d \mid z^{\,d})$$

This is exact Gibbs sampling from q_t — the model's joint distribution at noise level t. Repeated application converges to q_t regardless of initialisation.

Informed corrector. Instead of selecting d uniformly, select with probability proportional to the *surprise* at each position:

$$P(\text{select } d) \propto \exp(\beta \cdot [-\log q_t(z^d \mid z^{\,d})])$$

for temperature $\beta > 0$. High surprise (low log-probability $\to$ high $-\log q_t$) $\to$ high selection probability $\to$ more likely to be corrected. At $\beta = 0$ this reduces to the uniform corrector; as $\beta \to \infty$ it always selects the most surprised position.

Hollow transformer. Computing $q_t(z^d \mid z^{\,d})$ exactly for all d simultaneously requires one forward pass with each position masked in turn — $O(L)$ passes for a sequence of length L. The *hollow transformer* avoids this by using a modified attention mask that zeroes the diagonal (position i cannot attend to itself), enabling exact computation of all per-position conditionals $\{q_t(z^d \mid z^{\,d})\}_{d=1}^{L}$ in a single forward pass. This requires retraining.

Full algorithm:

```
# Phase 1: standard MDM sampling
z_0 ← standard_MDM_sampler(z_T)

# Phase 2: correction loop
for r = 1, ..., R:
    for d = 1, ..., L:   (or batched)
        # Compute surprise (needs hollow transformer for O(1) cost)
        surprise^d = -log q_t(z_0^d | z_0^{ d})
    # Select position to correct, proportional to exp(  · surprise)
    d* ← sample   exp(  · surprise)
    # Resample
    z_0^{d*} ← sample from q_t(z_0^{d*} | z_0^{ d*})
return z_0
```

7.2.3 Theoretical Contributions

Mixing time bound (Theorem 3). Let $\lambda > 0$ be the spectral gap of the Gibbs chain. After R correction steps:

$$\mathrm{KL}(q_0 \| p_R) \leq e^{-R\lambda} \cdot \mathrm{KL}(q_0 \| p_0)$$

where p_0 is the distribution at the start of correction and p_R is the distribution after R steps. The KL decays exponentially in R.

Informed corrector acceleration (Theorem 4). Let λ_u be the spectral gap for the uniform corrector and $\lambda_i(\beta)$ for the informed corrector at temperature β:

$$\lambda_i(\beta) \geq \lambda_u \quad \forall \beta \geq 0$$

with equality only when all positions have equal surprise. The informed corrector achieves the same KL reduction as the uniform corrector but in fewer steps. The gap $\lambda_i - \lambda_u$ grows with the variance of surprises across positions — exactly when the information profile is non-uniform, which is when the thesis predicts the largest benefit from confidence-guided remasking.

7.2.4 Confidence Signal

$-\log q_t(z^d \mid z^{\setminus d})$ — the negative log-probability of the current token at position d, given all other positions. This is the *exact* per-position conditional, the Tier 1 signal.

Computational cost without hollow transformer: $O(L)$ forward passes per correction step — prohibitive for long sequences.

With hollow transformer: $O(1)$ forward passes — efficient, but requires retraining the model with hollow attention.

Approximation for standard MDMs: Use the model's output probability $-\log p_\theta(z^d = z_0^d \mid z_0)$ (the negative log-probability that the current committed token z_0^d is the correct prediction) as a proxy. This is the Tier 3 margin signal.

7.2.5 Limitations

1. **Requires hollow transformer.** Not applicable to MDLM-OWT, LLaDA-8B, RemeDi, or any other standard pretrained MDM without retraining.
2. **Post-hoc correction only.** Informed Correctors is applied after standard sampling is complete, not integrated into the denoising loop. This means it cannot correct errors made by early unmasking decisions before those decisions propagate to later steps.
3. **Spectral gap λ can be very small.** For sequences with strong long-range dependencies (e.g., code), the Gibbs chain can mix very slowly, requiring an impractical number of correction steps.
4. **No connection to information profile $I(x)$ or E_{fact}.** The MCMC framework and the Riemann error framework are parallel but have not been formally connected.

7.2.6 Relation to This Thesis

Informed Correctors provides MCMC-theoretic corroboration for the thesis's main claim: Theorem 4 proves that confidence-guided correction is strictly more efficient than uniform correction, from a completely different theoretical perspective (spectral gap vs. Riemann error). The thesis can use this as supporting evidence.

The key design difference from the thesis: Informed Correctors requires exact per-position conditionals (Tier 1 signal, hollow transformer), while the thesis uses only forward-pass logits (Tier 3 signal, any pretrained MDM). The thesis's contribution is to show that Tier 3 signals are *sufficient* for the efficiency gain — you do not need the exact conditional to benefit from confidence-guided correction.

7.2.7 Study Questions

1. The informed corrector selects position $d^* \propto \exp(\beta \cdot [-\log q_t(z^d \mid z^{\ d})])$. At $\beta = 0$ this is uniform; at $\beta \to \infty$ it always picks the most surprised position. What is the optimal β? How would you tune it?
2. The hollow transformer computes all L per-position conditionals in a single forward pass by zeroing the attention diagonal. Why does zeroing the diagonal give $q_t(z^d \mid z^{\ d})$? Is this exact or approximate?
3. The spectral gap $\lambda_i \geq \lambda_u$ (informed $\geq$ uniform). The gap grows with the variance of surprises $\mathrm{Var}_d[-\log q_t^d]$. How does this variance relate to the information profile variance Σ^2?
4. Informed Correctors applies correction *after* the full generation is done. The thesis integrates remasking *into* the generation loop. What is the qualitative difference? Can you think of a scenario where post-hoc correction is better?

7.3 RemeDi (2025) — Learned Policy for Unmask and Remask Decisions

Full citation: Huang et al. (2025), "Don't Settle Too Early: Self-Reflective Remasking for Diffusion Language Models." arXiv:2509.23653. HuggingFace: `maple-research-lab/RemeDi-RL`, `maple-research-lab/RemeDi-Instruct`.

7.3.1 Summary

RemeDi introduces a fundamentally different approach to remasking: rather than using a heuristic or analytically derived confidence signal, it *learns* when to unmask and when to remask through a dedicated policy network (the Unmasking Policy Stream, UPS). The base token predictor (Token Prediction Stream, TPS) is unchanged; the UPS is a lightweight MLP trained on top of TPS hidden states, first with supervised learning (BCE on whether the current prediction is correct) and then refined with Group Relative Policy Optimisation (GRPO) reinforcement learning on a generation quality reward.

RemeDi represents the pinnacle of the fine-tuning approach: it has access to training signal not available at inference time and can learn globally optimal unmask/remask decisions that no training-free signal can match in principle. However, it requires a full training pipeline (TPS pre-training + UPS BCE training + GRPO RL) and the resulting policy is not portable to other model families.

7.3.2 Method Details

Architecture. TPS is a standard bidirectional transformer decoder trained with the MDM ELBO. UPS is a lightweight MLP applied to the TPS hidden states h^i at each position:

$$\psi^i = \sigma(W_3 \cdot \mathrm{ReLU}(W_2 \cdot \mathrm{ReLU}(W_1 \cdot h^i + b_1) + b_2) + b_3) \in [0, 1]$$

ψ^i is the UPS's estimate of "should token i be committed at this step?" — distinct from the TPS's estimate of "what token should be placed at position i?". This separation is the key architectural insight: the *what* and the *when* are different problems that can be learned independently.

Stage 1 — TPS pre-training. Standard MDLM ELBO training on a large text corpus. Produces a 9B-parameter bidirectional transformer competitive with LLaDA.

Stage 2 — UPS supervised pre-training. Given a training sequence x and a randomly masked version z_t (simulating a mid-generation state), train UPS to predict whether the TPS's top-1 prediction is correct:

$$y^i = \mathbf{1}[\hat{x}^i = x^i] \quad \text{where } \hat{x}^i = \arg\max_j p_\theta(x^i = j \mid z_t)$$

$$\mathcal{L}_{\text{UPS}} = -\sum_i \left[y^i \log \psi^i + (1 - y^i) \log(1 - \psi^i)\right]$$

After this stage, ψ^i is a calibrated estimator of the TPS's per-token accuracy. High ψ^i means the TPS's prediction at position i is likely correct; low ψ^i means it is likely wrong and the position should be remasked.

The paper proves (analogously to PRISM) that the BCE minimiser satisfies:

$$\psi^{i*} = P(\hat{x}^i = x^i \mid z_t)$$

i.e., UPS converges to the true conditional accuracy of the TPS.

Stage 3 — GRPO RL fine-tuning. The UPS is further refined by Group Relative Policy Optimisation with a reward signal r based on the quality of the fully generated sequence. GRPO compares multiple completions from the same prefix and updates the policy towards higher-reward ones, without requiring a separate reward model.

The RL stage teaches UPS to make *globally optimal* decisions — not just locally correct ones (which is what BCE training achieves). For example, BCE training might teach UPS to remask a locally uncertain token, but GRPO might discover that remasking that token always leads to worse final sequences (because the correction introduces a cascade of errors elsewhere).

Inference algorithm:

```
z_T = [m, m, ..., m]
for t = T, T-1, ..., 1:
    # Single forward pass through TPS+UPS
    p^i, ^i ← TPS+UPS(z_t)   for all positions i

    # Unmask confident masked positions
    for i with z_t^i = m and ^i >  _unmask:
        z_{t-1}^i ← argmax_j p^i_j   (or sample from p^i)

    # Remask uncertain committed positions
    for i with z_t^i  m and ^i <  _remask:
        z_{t-1}^i ← m

return z_0
```

The thresholds τ_{unmask} and τ_{remask} are hyperparameters (typically set to 0.5).

Stored decoding probability. RemeDi additionally stores $\psi^i(t^*)$ — the UPS score at the time token i was originally committed. In subsequent steps, if ψ^i drops significantly below $\psi^i(t^*)$, the token's confidence has deteriorated and it is a strong candidate for remasking. This gives RemeDi a form of memory across steps.

7.3.3 Theoretical Contributions

1. **UPS BCE minimiser converges to $P(\hat{x}^i = x^i \mid z_t)$.** Standard proper-scoring-rule argument (the BCE loss is a strictly proper scoring rule for probability estimation).
2. **GRPO convergence.** Standard GRPO convergence results apply: the RL fine-tuning converges to a local optimum of the expected reward. No global optimality guarantee.
3. **No sampling error bound.** RemeDi does not connect to E_{fact} or the information profile.

7.3.4 Confidence Signal

$\psi^i \in [0, 1]$ — a learned, calibrated estimate of $P(\text{TPS prediction at position } i \text{ is correct} \mid z_t)$.

Properties: - Calibrated across noise levels (BCE training on diverse t). - Accounts for global context via the TPS hidden state h^i. - Refined towards globally optimal decisions by GRPO. - **Not portable:** must be retrained for each new base model.

7.3.5 Limitations

1. **Three-stage training pipeline.** Requires TPS pre-training + UPS BCE + GRPO RL. The RL stage in particular is sensitive to hyperparameters and can be unstable at 9B scale.
2. **Not portable.** UPS is trained for a specific TPS architecture and corpus; it cannot be transferred to MDLM-OWT or LLaDA without retraining.
3. **No connection to optimal sampling theory.** The learned policy is not shown to be optimal in any theoretical sense; it is empirically motivated by the RL reward.
4. **Evaluation scope.** RemeDi is evaluated primarily on instruction-following tasks; unconditional quality metrics (MAUVE, generative perplexity) are underreported.

7.3.6 Relation to This Thesis

RemeDi is the empirical upper bound for the thesis. The RL-trained UPS can in principle learn any remasking policy, while the thesis is constrained to training-free signals. The thesis's key empirical question is: how much of RemeDi's quality advantage can be recovered by the thesis's principled training-free strategy? If the gap is small, training is unnecessary; if large, the gap is explained by the difference between Tier 2 (learned) and Tier 3 (heuristic) signals — which is itself a contribution.

7.3.7 Study Questions

1. RemeDi's UPS is trained with BCE on whether the TPS prediction is correct. The BCE minimiser converges to $P(\hat{x}^i = x^i \mid z_t)$. Is this the same as $I^i(x)$? When do they agree and when do they diverge?
2. RemeDi uses GRPO RL to refine the UPS beyond BCE training. What does GRPO add that BCE training cannot provide? Give an example of a decision that would be suboptimal under BCE but optimal under GRPO.
3. The RemeDi-RL model on HuggingFace is 8B parameters with an 8B tokenizer. Our MDLM-OWT baseline is 130M. If you wanted to compare RemeDi's remasking strategy (not model scale) to the thesis's strategy, what experimental design would be fair?

4. RemeDi stores $\psi^i(t^*)$ — the UPS score at the time token i was committed. This gives it memory across steps. Does the thesis's entropy-based remasking have an implicit memory? What would it mean to add explicit memory to a Tier 3 strategy?

7.4 PRISM (2025) — Provable Quality Head with a Formal Guarantee

Full citation: Kim et al. (2025), "Fine-Tuning Masked Diffusion for Provable Self-Correction." arXiv:2510.01384. GitHub: `SeunggeunKimkr/PRISM`.

7.4.1 Summary

PRISM trains a lightweight MLP quality head g_ϕ on top of a frozen pretrained MDM to predict, for each token position, the probability that the model's current prediction is correct. This is the simplest possible approach to learning a confidence signal: the base model is completely unchanged, and only a small MLP (a few thousand parameters) is trained.

PRISM's theoretical contribution is a formal convergence guarantee: the BCE minimiser of g_ϕ's training objective converges to the true Bayes-optimal conditional accuracy $P(x^i = \hat{x}^i \mid y \oplus m^i)$, where $y \oplus m^i$ is the input sequence with position i additionally masked. This is the strongest theoretical result for any confidence signal in the MDM remasking literature.

7.4.2 Method Details

Architecture. g_ϕ is a lightweight MLP (2–3 hidden layers, ReLU activations) applied to the frozen hidden states h^i of the base MDM:

$$g_\phi(y)^i = \sigma(W_k \mathrm{ReLU}(\cdots W_2 \mathrm{ReLU}(W_1 h^i + b_1) + b_2 \cdots) + b_k) \in [0, 1]$$

The base model p_θ is completely frozen; only $\{W_1, \ldots, W_k, b_1, \ldots, b_k\}$ are trained.

Training objective. For a training pair (x, y) where y is a partially masked version of x and $\hat{x}^i = \arg\max_j p_\theta(x^i = j \mid y)$ is the base model's top-1 prediction:

$$\mathcal{L}(\phi) = -\mathbb{E}_{x,y}\left[\sum_i \mathbf{1}[x^i = \hat{x}^i] \log g_\phi(y)^i + \mathbf{1}[x^i \neq \hat{x}^i] \log(1 - g_\phi(y)^i)\right]$$

This is a standard binary cross-entropy loss where the label is 1 if the base model's prediction is correct and 0 otherwise.

Theoretical guarantee (Proposition 1 of PRISM). The minimiser ϕ^* over the class of all measurable functions satisfies:

$$g_{\phi^*}(y)^i = P(x^i = \hat{x}^i \mid y \oplus m^i)$$

where $y \oplus m^i$ is the sequence y with position i additionally masked. This is the Bayes-optimal prediction of the base model's accuracy at position i, given the information available in y minus position i itself.

Proof sketch. The BCE loss is a strictly proper scoring rule: for any event A, $\mathbb{E}[-y_A \log q - (1 - y_A)\log(1-q)]$ is minimised at $q = P(A)$. Applying this to $A = \{x^i = \hat{x}^i\}$ with the conditioning variable being all information except y^i (hence $y \oplus m^i$) gives the result.

Inference algorithm. At generation step t:

```
# One pass through frozen base model: get logits and hidden states
p^i, h^i ← p_ (z_t)   for all i
# One pass through quality head: get confidence scores
g^i ← g_ (h^i)   for all i (free: just an MLP)
# Unmask top-K most confident masked positions
unmask_set ← top-K {i : z_t^i = m} by g^i
z_{t-1}^i ← argmax p^i   for i in unmask_set
# Optionally remask low-confidence committed positions
for i with z_t^i  m and g^i <  :
    z_{t-1}^i ← m
```

Evaluation. PRISM is evaluated on three tasks: - **Sudoku:** discrete constraint satisfaction — PRISM dramatically improves completion accuracy. - **MDLM-OWT (170M):** unconditional text generation — improves generative perplexity and MAUVE. *(Formal definitions of generative perplexity and MAUVE: Appendix A.8.1–A.8.2.)* - **LLaDA-8B (MBPP code):** code generation — improves pass@1 on the MBPP benchmark, demonstrating scalability.

7.4.3 Theoretical Contributions

1. **Bayes-optimality of the BCE minimiser.** The formal convergence result is the strongest guarantee for any confidence signal in the MDM remasking literature. It shows that g_ϕ is not just a heuristic but the optimal predictor of base model accuracy given the available information.
2. **Connection to PRISM's name.** The quality head predicts whether the model's prediction "passes" (is correct) — hence Provable Self-correction via Inference-time Self-Monitoring (PRISM).

7.4.4 Confidence Signal

$g_\phi(y)^i \in [0,1]$ — the Bayes-optimal estimate of $P(\hat{x}^i = x^i \mid y \oplus m^i)$.

Properties: - Theoretically optimal under the BCE minimisation criterion. - Calibrated across noise levels (training covers all t). - Negligible computational overhead (tiny MLP on frozen hidden states). - **Requires training:** a new g_ϕ for each base model.

7.4.5 Limitations

1. **Training required.** A new g_ϕ must be trained for each base model. Requires labelled data (parallel clean/masked sequences) and compute.
2. **LLaDA-8B adapter not public.** As of early 2026, the quality head trained on LLaDA-8B is available only to lab collaborators.
3. **Guarantee is for calibration, not sampling quality.** Theorem 1 proves that g_ϕ converges to the true conditional accuracy. It does not prove that using g_ϕ as a sampling guide produces sequences from π or reduces E_{fact}.

4. **OOD calibration.** g_ϕ is trained on a specific corpus; calibration may degrade for prompts far from the training distribution.

7.4.6 Relation to This Thesis

PRISM's BCE convergence theorem provides the formal justification for the thesis's central assumption. The thesis assumes that a training-free signal c^i is a *consistent estimator* of $H(x^i \mid \text{context})$ (Tier 3 signal behaves like Tier 1). PRISM shows that with a small amount of training, exact convergence to the true conditional is achievable (Tier 2). The thesis bridges the gap: it provides formal conditions on c^i (consistency) under which the Tier 3 signal is *sufficient* to achieve an E_{fact} reduction comparable to what PRISM would achieve. If these conditions hold for max-prob/entropy/margin in practice (which the experiments test), then retraining is unnecessary.

7.4.7 Study Questions

1. PRISM's Proposition 1 says $g^i_{\phi^*} = P(x^i = \hat{x}^i \mid y \oplus m^i)$. The conditioning variable is $y \oplus m^i$ (sequence with position i also masked). Why is it conditioned on $y \oplus m^i$ and not on y itself?
2. PRISM dramatically improves Sudoku completion accuracy. Why would a discrete diffusion model with PRISM be particularly effective on constraint satisfaction tasks? What does the confidence score capture that matters for Sudoku?
3. PRISM's quality head g_ϕ is a tiny MLP on frozen hidden states. Could you use a different architecture (e.g., a linear probe, or attention over all positions)? What would be gained or lost?
4. PRISM and RemeDi both use BCE loss and both converge to the true conditional. The difference is that RemeDi also adds RL fine-tuning. Suppose PRISM also added an RL stage on top of g_ϕ. Would this give the same result as RemeDi? What would be different?

8 Part VI — Cross-Paper Comparison

8.1 Chronological Overview and Intellectual Lineage

The twelve papers form two parallel development threads that the thesis unifies:

Thread A — Non-autoregressive generation and remasking practice:

Mask-Predict (2019) → MDLM (2024) → LLaDA (2025) → ReMDM (2025) → RemeDi (2025), PRISM (2025)

This thread develops increasingly powerful remasking strategies, starting from the pure heuristic (Mask-Predict) and progressing to learned approaches (PRISM, RemeDi). Quality improves at each step, but theoretical understanding lags behind.

Thread B — Discrete diffusion theory:

D3PM (2021) → SEDD (2024) → DFM (2024) → Lavenant & Zanella (2025) → EB-Sampler (2025)

This thread develops the theoretical framework for understanding and improving MDM sampling, culminating in the optimal unmasking schedule. It is rigorous but stops short of the remasking direction.

The thesis. Sits at the confluence of both threads: it brings the rigour of Thread B to the practice of Thread A. The main theorem characterises confidence-guided remasking within the Lavenant & Zanella E_{fact} framework, connecting the optimal Tier 3 signal (entropy, from EB-Sampler) to the convergence guarantees of Thread B.

8.2 Comprehensive Comparison Table

#	Paper	Year	Backbone	Confidence Signal	Tier	Training?	Remasking?	Theoretical Guarantee	Key Metric	Dataset	Approx. Params
1	Mask-Predict	2019	BERT-scale	Max-prob	3	Pre-train	Yes (heuristic)	None	BLEU	WMT	~200M
2	D3PM	2021	Custom	None	—	Pre-train	No	VLB decomp.	NLL	Text8	~100M
3	SEDD	2024	GPT-2 scale	Score ratio	1	Pre-train	PC corrector	Consistency + PC conv.	Perplexity	OWT	~300M
4	MD4 / MDLM	2024	Custom	None	—	Pre-train	No	ELBO simplif.	Perplexity	OWT, LM1B	130M

#	Paper	Year	Backbone	Confidence Signal	Tier	Training?	Remasking?	Theoretical Guarantee	Key Metric	Dataset	Approx. Params
5	DFM	2024	Any MDM	Velocity entropy	3	Pre-train	Corrector	Path-integral KL	Perplexity	OWT, audio	Any
6	Lav. & Zan.	2025	Any MDM	N/A (analysis)	—	No	Open problem	KL $= E_l + E_f$	N/A	N/A	—
7	EB-Sampler	2025	Any MDM	Entropy $H(p_\theta)$	3	No	No	E_f Riemann bd.	PPL, speed	OWT	Any
8	ReMDM	2025	MDLM-OWT	Entropy (opt.)	3	No	Yes (principled)	KL monotone in R	Perplexity	OWT	130M
9	Inf. Corr.	2025	Hollow transf.	Log-margin (exact)	1	Retrain	Yes (MCMC)	MCMC mixing time	PPL, diversity	Text8, OWT	Custom
10	LLaDA	2025	8B transformer	None	—	Pre-train	Uniform	None (empirical)	Benchmarks	2.3T tokens	8B
11	RemeDi	2025	TPS+UPS (9B)	Learned ψ^i	2	Yes (RL)	Yes (learned)	UPS BCE conv.	Instruction acc.	Custom	9B
12	PRISM	2025	LLaDA-8B	Learned g_ϕ	2	Yes (head)	Yes (guided)	BCE $\to$ true cond.	Gen. quality	OWT, MBPP	8B + MLP

Tier: 1 = exact posterior ratio; 2 = learned approximation; 3 = heuristic training-free.

8.3 Dimension 1: Training Requirements

Approach	Papers	Training required	Portable to new model?
Pre-train only (no special remasking)	D3PM, MDLM, LLaDA	Base model pre-training only	Yes

Approach	Papers	Training required	Portable to new model?
Training-free remasking	Mask-Predict, ReMDM, EB-Sampler, DFM, **Thesis**	None beyond pre-training	Yes
Fine-tuned head	PRISM	Small MLP on frozen model	No (per-model)
Fine-tuned policy (RL)	RemeDi	Full TPS+UPS+RL pipeline	No (per-model)
Architecture-retrained	Informed Correctors, SEDD	Full retraining with hollow attention or score objective	No (per-architecture)

The thesis occupies the training-free cell: it works with any pretrained MDM (MDLM-OWT, LLaDA-8B, RemeDi's TPS) using only the forward-pass logits. Its portability is its primary practical advantage over PRISM and RemeDi, and is what makes it relevant for any newly released MDM backbone.

8.4 Dimension 2: Theoretical Framework and Guarantee Strength

Paper	Framework	What is proved	Quantitative bound?
D3PM	Markov chain VLB	VLB decomposition	Yes (VLB lower bound)
SEDD	Score entropy	Score consistency + PC convergence	Yes (convergence rate)
DFM	Flow matching / probability paths	Objective equivalence, optimal path	Partial
Lavenant & Zanella	Riemann approximation of $I(x)$	KL = $E_l + E_f$; optimal schedule	Yes (explicit bound)
EB-Sampler	Riemann approximation of $I(x)$	E_f minimisation; optimality	Yes (explicit bound)
ReMDM	Markov chain monotonicity	KL monotone in R	Yes (monotonicity)
Informed Correctors	MCMC spectral gap	Exponential KL decay; informed > uniform	Yes (spectral bound)
PRISM	Proper scoring rules	BCE minimiser = true conditional	Yes (Bayes-optimality)
Mask-Predict, LLaDA	None	None	No
RemeDi	Proper scoring rules	UPS BCE $\rightarrow$ true conditional	Yes (for calibration only)

The thesis targets a result in the Lavenant & Zanella framework: a quantitative bound on the E_{fact} reduction achieved by confidence-guided remasking. This places it in the strongest theoretical quadrant (quantitative, information-theoretic) and is the gap no existing paper fills.

8.5 Dimension 3: The Confidence Signal Hierarchy

All remasking papers must answer: *which tokens should be remasked?* Their answers span three tiers:

Tier 1 — True posterior (exact, expensive). $I^i(x) = H(x^i \mid x^{\ i})$ — the conditional entropy of position i given all other positions. This is the gold standard: it exactly measures how much each position would benefit from being re-predicted in perfect context. Computing it requires either knowing x (unavailable) or $O(L)$ forward passes (expensive). Informed Correctors and SEDD operate at this tier (with the hollow transformer approximation).

Tier 2 — Learned approximation (calibrated, requires training). PRISM's g_ϕ^i and RemeDi's ψ^i approximate Tier 1 using a learned model. Both come with formal convergence guarantees (BCE minimiser = true conditional). The trade-off: they are calibrated and accurate but require training data and compute for each new backbone.

Tier 3 — Heuristic training-free (cheap, approximate, no guarantee). Three signals computed directly from $p_\theta(x^i \mid z_t)$: - **Max-probability:** $c^i = \max_j p_\theta^j$. Used by Mask-Predict. Intuitively: how confident is the model? But max-probability conflates unimodal (easy) with multi-modal (hard) distributions. - **Entropy:** $c^i = -H(p_\theta(x^i \mid z_t))$. Used by ReMDM and EB-Sampler. Directly approximates $I^i(x)$; theoretically the most principled of the three. - **Margin:** $c^i = p_\theta^{(1)} - p_\theta^{(2)}$. The gap between the top-2 predictions. Sensitive to multi-modal uncertainty that entropy may miss.

The thesis provides the first formal analysis of when and why Tier 3 signals are sufficient: under a consistency condition (the signal is an unbiased estimator of $I^i(x)$ in expectation), the E_{fact} reduction of the confidence-guided corrector equals that of the Tier 1 signal up to a variance term.

8.6 Dimension 4: Inference Compute Cost

Paper	NFE per token	Notes
MDLM (T steps)	T total, each unmasks L/T tokens	$T = 64$–1024 typically
EB-Sampler	$\approx T$ (adaptive)	$T \approx I(x)/\varepsilon$; often fewer than fixed-T MDLM
ReMDM (T steps, R corrections)	$T \times (R+1)$	Linear in R; $R = 1$–10 practical
PRISM	T + negligible (MLP)	Quality head overhead $\ll$ base model
RemeDi	T (single pass per step)	UPS adds $\sim 0.1\%$ overhead
Informed Correctors	$T + R \times L$ (without hollow transf.)	$O(L)$ per correction step; impractical
Informed Correctors (hollow)	$T + R$	With hollow transformer: same as ReMDM

Thesis (threshold/top-k)	$T \times (1 + r)$, $r \leq 0.2$ typically	Only high-entropy tokens are remasked

The thesis's strategies have a compute cost between standard MDLM ($r = 0$) and ReMDM ($r = R$). For small τ (tight threshold), few tokens are remasked and $r \approx 0.05$–0.1, giving a 5–10% overhead. This is the most compute-efficient remasking strategy in the comparison.

8.7 Dimension 5: Empirical Results and Evaluation Scope

Paper	Metric	Best result	Scale
MDLM	Perplexity (OWT)	~26–27	130M
SEDD	Perplexity (OWT)	~25	300M
LLaDA	MMLU, GSM8K, etc.	Competitive with LLaMA 3 8B	8B
ReMDM	Gen. perplexity (OWT)	15–20% reduction vs. MDLM	130M
RemeDi	Instruction acc.	SotA among MDMs	9B
PRISM	MBPP pass@1	+5–10% vs. base LLaDA	8B
Informed Corr.	PPL + diversity	+10% vs. no correction	150M (hollow)
EB-Sampler	PPL at fixed NFE	2–3× speedup or +5% PPL	130M

The thesis should target: (1) matching or exceeding ReMDM's 15–20% generative perplexity reduction on MDLM-OWT with lower compute overhead; (2) demonstrating a 5–10% improvement on LLaDA-8B on downstream tasks (LAMBADA, HellaSwag); (3) showing the empirical benefit scales with the variance of the information profile Σ^2 (non-uniform sequences benefit more).

8.8 Dimension 6: Connection to the Information Profile $I(x)$

Paper	Uses $I(x)$?	How?
D3PM, MDLM, LLaDA	No	Uniform random unmasking
Mask-Predict	Implicitly	Max-prob as proxy for I^i
SEDD	Implicitly	Score entropy $\approx$ log-ratio proxy
DFM	Implicitly	Velocity entropy $= H(p_\theta^i)$
Lavenant & Zanella	**Explicitly**	E_{fact} = Riemann error of $\int I(x)\,dt$
EB-Sampler	**Explicitly**	Unmask in ascending order of $\hat{I}^i$

Paper	Uses $I(x)$?	How?
ReMDM	No	Remasking prob. based on α_t, not I^i
Informed Correctors	No	Surprise signal $\neq I^i$ (it is $-\log q_t^d$)
RemeDi, PRISM	No	Accuracy proxy, not information proxy
Thesis	**Explicitly**	Remask in descending order of $\hat{I}^i$; derive E_f reduction

The thesis is the only paper that connects *remasking* to the information profile. Lavenant & Zanella and EB-Sampler connect *unmasking* to $I(x)$; the thesis completes the picture.

8.9 The Open Gap: Where the Thesis Fits

After reviewing all twelve papers, the landscape of MDM sampling theory has one conspicuous gap:

Direction	Optimal algorithm	Theoretical bound	Confidence signal analysis
Unmasking	EB-Sampler (2025)	Lavenant & Zanella (2025)	Implied by $I^i(x) \approx H(p_\theta^i)$
Remasking	???	???	???

The thesis fills all three cells of the remasking row. Its main result:

> **Theorem (thesis, informal).** Let $\mathcal{A}_\tau$ be the MDM sampler that, at each step, (1) unmasks tokens in ascending order of $\hat{I}^i$ (EB-Sampler rule) and (2) remasks committed tokens with $\hat{I}^i > \tau$ (confidence-guided corrector). If $\hat{I}^i$ is a consistent estimator of $I^i(x)$ in the sense that $\mathbb{E}[\hat{I}^i \mid x] = I^i(x) + \delta$ with $|\delta| \leq \epsilon$, then:
>
> $$E_{\text{fact}}(\mathcal{A}_\tau) \leq E_{\text{fact}}(\text{EB-Sampler}) - \Delta(\tau, I, \epsilon)$$
>
> where $\Delta(\tau, I, \epsilon) > 0$ whenever the information profile is non-uniform ($\Sigma^2 > 0$), the threshold τ is chosen to target the top-k fraction of positions by I^i, and ϵ is small relative to $I(x)/L$.

This result: - **Extends Lavenant & Zanella** to the remasking direction. - **Explains ReMDM's empirical success** (entropy-weighted remasking reduces E_{fact}) and the gap over uniform remasking (uniform ignores I^i, so $\Delta = 0$ for the uniform corrector). - **Provides a formal justification for PRISM and RemeDi**: their learned signals are consistent estimators of I^i by design (BCE minimiser = true conditional); the thesis's condition is automatically satisfied. - **Connects all six remasking papers** (Mask-Predict, ReMDM, Informed Correctors, RemeDi, PRISM, EB-Sampler) within a single framework.

9 Part VII — Research Directions: Principled Rescheduling

The term *rescheduling* refers to the combined problem of deciding, at each inference step: (a) *which* tokens to unmask, (b) *how many*, and (c) *which committed tokens to remask*. The EB-Sampler solves (a) and (b); the thesis adds (c). But the full rescheduling problem has many open directions. This part surveys them systematically.

Context from our experiments. Our step sweep (T {128, 256, 512, 1000}) reveals phenomena that motivate each direction: - MDLM MAUVE peaks at T=256 then drops → diversity window; optimal T is not ∞ - remdm-conf MAUVE collapses at high T → confidence overconfidence problem - remdm-loop MAUVE improves monotonically → loop remasking is robust but slow - gen_ppl and MAUVE decouple sharply → mode-seeking vs. coverage tension *(formal definitions and the gen_ppl decomposition into $H(p_\theta) + D_{KL}(p_\theta \| q_{GPT\text{-}2})$: Appendix A.8.1–A.8.2)*

9.1 R1: Unified EB-Sampler + Confidence-Guided Remasking

9.1.1 The Gap

The EB-Sampler (optimal unmasking schedule) and ReMDM (principled remasking) were developed independently and have never been combined. Their interaction is unknown: does entropy-budget unmasking *amplify* or *suppress* the benefit of remasking?

9.1.2 Formal Setup

Let $\mathcal{A}_{\varepsilon,\tau}$ be the sampler that: (1) uses the EB-Sampler with budget ε for unmasking, and (2) remasks committed tokens with $\hat{I}^i > \tau$ after each unmasking step. Define:

$$E_{\text{fact}}(\mathcal{A}_{\varepsilon,\tau}) = E_{\text{fact}}^{\text{unmask}}(\varepsilon) + E_{\text{fact}}^{\text{remask}}(\tau) + E_{\text{interaction}}(\varepsilon, \tau)$$

The first two terms are bounded by the EB-Sampler theorem and the thesis theorem, respectively. The interaction term $E_{\text{interaction}}$ is unknown.

Hypothesis. The interaction is *negative* (beneficial): after remasking corrects high-entropy committed tokens, the EB-Sampler's entropy estimates become more accurate for the next unmasking step, reducing $E_{\text{fact}}^{\text{unmask}}$ below its standalone value. Remasking and unmasking are *synergistic*.

9.1.3 Key Questions

- What is the optimal relationship between ε and τ?
  - Proposal: $\tau = \varepsilon$ — remask exactly the tokens the EB-Sampler would have committed last, i.e., the ones that barely fit in the entropy budget.
  - Intuition: tokens just above the budget threshold are the ones where the unmasking decision is most uncertain. Remasking them allows the model to re-evaluate in richer context.
- Does the order of operations matter? (unmask then remask, vs. remask then unmask, vs. interleaved)?
- Can the combined sampler be shown to achieve a tighter E_{fact} bound than either strategy alone?

9.1.4 Implementation

Modify `external/remdm/main.py` to replace the uniform unmasking step with the EB-Sampler step. The entropy budget ε and remasking threshold τ become two hyperparameters. Run the step sweep for combinations of (ε, τ).

9.2 R2: Calibrated Confidence — Temperature Scaling for Remasking

9.2.1 The Problem

Our remdm-conf MAUVE collapse at high T is a **calibration failure**: the model's softmax distribution over tokens becomes overconfident at low masking rates (few masked positions $\to$ rich context $\to$ sharply peaked predictions). Overconfident predictions produce low entropy $\widehat{I}^i$, so high-entropy positions (which need correction) are *underdetected*, and the model locks in poor early choices.

Formal statement. Define calibration error at noise level t as:

$$\mathrm{ECE}(t) = \mathbb{E}\Big[\Big|\widehat{I}^i(z_t) - I^i(x)\Big|\Big]$$

Hypothesis: $\mathrm{ECE}(t)$ increases as $t \to 0$ (low masking rate) because the model's entropy estimate degrades when context is rich but the model is still uncertain about some positions for non-local reasons.

9.2.2 Temperature Scaling Solution

Replace the entropy estimate with a temperature-scaled version:

$$\widehat{I}^i_\tau(z_t) = H\left(\mathrm{softmax}\left(\frac{\ell^i}{\tau(t)}\right)\right)$$

where ℓ^i are the raw logits and $\tau(t) > 1$ inflates entropy estimates. Three schedule options:

1. **Fixed:** $\tau(t) = \tau_0$ (constant inflation)
2. **Annealed:** $\tau(t) = 1 + (\tau_0 - 1) \cdot (1 - \alpha_t)$ where α_t is the survival probability. Since $\alpha_t \approx 0$ at the start (t T, high masking rate) and $\alpha_t \approx 1$ at the end (t 1, low masking rate), this gives $\tau \approx \tau_0$ early (high temperature, diverse) and $\tau \approx 1$ late (no softening, commit to best token).
3. **Data-adaptive:** $\tau(t)$ chosen so $\mathbb{E}[\widehat{I}^i_\tau] = \mathbb{E}[I^i]$ using a calibration set (requires a small labelled corpus)

The annealed schedule is the most theoretically motivated: it corrects the systematic overconfidence that appears as more tokens are committed (lower masking rate $\to$ lower α_t).

9.2.3 Connection to the Diversity Window

The diversity window (MDLM MAUVE peak at T=256) may itself be a calibration artefact. At T=1000, the model makes 1000 unmasking decisions, many at very low masking rates where $\widehat{I}^i$ is underestimated. The model prematurely commits low-entropy tokens, accumulating commitment errors. Temperature scaling that inflates $\widehat{I}^i$ at low t would delay commitment, potentially eliminating the diversity window — or shifting it to higher T.

Testable prediction: if temperature scaling eliminates (or substantially reduces) the remdm-conf MAUVE collapse at T=1000, this result is *consistent with* overconfidence being a primary cause — but does not prove it. If collapse persists across a range of τ_0, the cause is more likely structural (mode-seeking is intrinsic to the strategy, not a calibration artefact), and a fundamentally different remasking policy would be needed.

9.2.4 Implementation

No retraining required. Add a `--temperature_schedule` flag to `scripts/remedi_eval.py` and the ReMDM inference loop. Calibration curves can be estimated by comparing $\hat{I}^i$ to empirical accuracy on a held-out corpus.

9.3 R3: The Remasking-Schedule Co-design Problem

9.3.1 The Problem

The current framework treats the noise schedule α_t (set during training) and the inference-time rescheduling strategy as independent. But they are coupled: the optimal remasking threshold τ depends on α_t, and conversely, the optimal α_t depends on how much remasking will be used at inference.

Formal observation. The EB-Sampler bound $E_{\text{fact}} \leq \varepsilon \cdot \max_i \hat{I}^i$ depends on $\hat{I}^i$, which depends on α_t through the noise level. A noise schedule that makes $\hat{I}^i$ more uniform across positions (lower Σ^2) will reduce E_{fact} for the same entropy budget ε.

9.3.2 Research Questions

1. **Optimal schedule given remasking.** The Lavenant & Zanella bound uses a fixed α_t (cosine or linear). If remasking is used, the effective number of "independent" decisions per step changes. Derive the optimal α_t that minimises E_{fact} given that a confidence-guided corrector will be applied at each step.
2. **Curriculum training.** Train the model with a noise schedule that concentrates on the masking rates where the remasking threshold τ will be most active. For example, if τ is set to remask positions with $\hat{I}^i > 0.5$ nats, then training should over-sample α_t corresponding to this entropy level.
3. **Learned vs. fixed schedule.** The noise schedule is a fixed cosine curve in all current MDMs. Could a *learned* schedule (optimised end-to-end with the remasking strategy) outperform the cosine schedule? This is the MDM analogue of continuous diffusion's noise schedule optimisation (e.g., Kingma et al., 2021).

9.3.3 Difficulty

Noise schedule optimisation requires retraining, making this a combined training + inference problem. However, the co-design insight (that schedule and rescheduling strategy should be optimised jointly) can be stated and verified empirically by testing multiple schedules with the same rescheduling strategy.

9.4 R4: Block-Level Rescheduling for Large Models

9.4.1 The Problem

All current rescheduling work (EB-Sampler, ReMDM, thesis) operates at the *token* level: one decision per token per step. For 8B-scale models with sequence lengths of 2048 tokens, token-level rescheduling over T=256 steps requires $256 \times 2048 = 524{,}288$ individual decisions per sequence. The overhead of sorting, thresholding, and remasking at token granularity can be non-negligible.

Block-wise generation (used by LLaDA and RemeDi) groups B tokens into a block and generates all B tokens together. This is computationally efficient but uses a fixed block size (typically $B = 32$), ignoring the information profile within each block.

9.4.2 Formal Proposal: Hierarchical EB-Sampler

Define a two-level rescheduling strategy: - **Block level:** use the EB-Sampler to decide *which blocks* to unmask at each step, based on the block's average entropy $\bar{I}^b = \frac{1}{B}\sum_{i \in b} \hat{I}^i$. - **Token level (within committed blocks):** use confidence-guided remasking to revisit high-entropy tokens within recently committed blocks.

Block-level entropy budget:

$$\varepsilon_B = B \cdot \varepsilon_{\text{token}}$$

Unmask all blocks whose average entropy $\bar{I}^b \leq \varepsilon_B$ per step. This preserves the EB-Sampler's structure but operates at block granularity.

Interaction between levels: Token-level remasking within a block changes the block's average entropy. Should the block-level schedule be updated after within-block corrections?

9.4.3 Why This Matters for the Thesis

If the block-level EB-Sampler achieves comparable E_{fact} reduction to the token-level EB-Sampler, then the thesis's contribution scales to 8B models without the $O(L)$ overhead of per-token sorting. The key quantity to bound is the error introduced by replacing $\hat{I}^i$ with $\bar{I}^b$ — a Jensen's inequality argument.

9.5 R5: Multi-Pass Rescheduling and Convergence

9.5.1 The Problem

The current framework applies one remasking pass per denoising step: unmask (EB-Sampler), then remask (thesis strategy). What is the benefit of applying *multiple* remasking passes per step, and does the benefit per pass diminish?

9.5.2 Formal Framework

Define the k-pass sampler $\mathcal{A}^{(k)}_{\varepsilon,\tau}$:

```
for each denoising step t → t-1:
    z' ← EB-Sampler(z_t, )           # unmask
    for k = 1, ..., K:
        z' ← remask_high_entropy(z', )  # remask
```

```
        z' ← EB-Sampler(z',  _k)       # re-unmask (with reduced budget  _k)
z_{t-1} ← z'
```

The budget ε_k for re-unmasking after the k-th remasking pass should decrease geometrically: $\varepsilon_k = \varepsilon/r^k$ for $r > 1$ (tighter budget each pass, ensuring convergence).

Question: Is there a formal convergence guarantee analogous to ReMDM's monotone improvement theorem? The ReMDM proof uses the data-processing inequality on the Markov chain; the same argument should apply to the combined EB+remask pass.

Claim: Under the assumption $E_{\text{learn}} = 0$:

$$E_{\text{fact}}(\mathcal{A}^{(k+1)}) \leq E_{\text{fact}}(\mathcal{A}^{(k)}) \quad \forall k$$

with the improvement per pass bounded by $\Delta_k \propto \varepsilon_k \cdot \tau$.

9.5.3 Relationship to Inference-Time Compute Scaling

If the bound on E_{fact} per pass decreases geometrically, then:

$$E_{\text{fact}}(\mathcal{A}^{(K)}) \leq E_{\text{fact}}(\mathcal{A}^{(0)}) \cdot r^{-K}$$

This is an *inference-time scaling law*: quality improves as $O(\log K)$ in the number of passes. A log-linear quality-compute curve is a practical finding that practitioners can use to decide how much inference compute to allocate.

9.6 R6: Rescheduling for Conditional Generation

9.6.1 The Problem

All existing rescheduling work (EB-Sampler, ReMDM, thesis) is evaluated on *unconditional* generation (no prompt). For practical use cases, the model generates a response y conditioned on a prompt x. The information profile $I^i(y)$ of the response depends on the prompt: an easy prompt (e.g., “Write: hello”) leads to a low-entropy response; a hard prompt (e.g., “Write a Shakespearean sonnet”) leads to a high-entropy response.

Formal setup. Conditional generation with masking:

$$q_{\text{cond}}(z_t \mid x, y) = \prod_i \text{Cat}(z_t^i;\, \alpha_t e_{y^i} + (1 - \alpha_t) e_m)$$

The information profile is now:

$$I^i_{\text{cond}}(y \mid x) = H(y^i \mid y^{\ i}, x)$$

which includes the prompt as conditioning. The EB-Sampler and remasking strategies should use I^i_{cond} rather than I^i.

9.6.2 Research Questions

1. **Prompt-adaptive entropy budget.** For easy prompts (low I^i_{cond}), fewer steps are needed. For hard prompts, more. Design an adaptive budget $\varepsilon(x)$ that scales with the prompt's estimated difficulty. How would you estimate difficulty from x alone (before generating y)?

2. **Conditional calibration.** Does the model's entropy estimate $\hat{I}^i(z_t, x)$ remain well-calibrated across prompts of different difficulties? A calibration analysis on LAMBADA or HellaSwag (where the prompt is the question and the response is the answer) would test this.

3. **Remasking for instruction following.** In LLaDA-8B-Instruct, the response must follow specific instructions (format, content). Does confidence-guided remasking improve instruction adherence? Hypothesis: instruction-critical tokens (e.g., format tokens like "1." or " "'python") have systematically lower confidence early in generation and benefit most from remasking.

9.7 R7: Information Profile Estimation Beyond Entropy

9.7.1 The Problem

The EB-Sampler and all Tier 3 signals use $\hat{I}^i = H(p_\theta(x^i \mid z_t))$ as a proxy for $I^i(x) = H(x^i \mid x^{\,i})$. This approximation has two known failure modes:

1. **Context sparsity.** Early in generation (high masking rate), z_t has few committed tokens. The model predicts x^i in low-context, making $\hat{I}^i$ a noisy estimate of $I^i(x)$.

2. **Multimodal uncertainty.** If $p_\theta(x^i \mid z_t)$ is bimodal (e.g., 50% "cat", 50% "dog"), the entropy is high but the correct token may be fully determined by context the model hasn't seen yet (masked positions). $\hat{I}^i$ overestimates $I^i(x)$ in this case.

9.7.2 Alternative Signals

Mutual information proxy:

$$\hat{I}^i_{\text{MI}}(z_t) = I(x^i; x_{\mathcal{M}_t}) \approx H(p_\theta(x^i \mid z_t)) - H(p_\theta(x^i \mid z_t^{\text{full}}))$$

where z_t^{full} is the sequence with all currently masked positions filled in with their MAP estimate. This estimates how much the masked positions would change the prediction at position i — a second-order uncertainty signal. Cost: one extra forward pass per step (fill in MAPs, then re-evaluate entropy).

Attention-based proxy:

$$\hat{I}^i_{\text{attn}}(z_t) = \sum_{j \in \mathcal{M}_t} a_{ij} \cdot \hat{I}^j(z_t)$$

where a_{ij} is the attention weight from position i to masked position j. This estimates how much position i would "care about" the currently masked positions if they were revealed. High $\hat{I}^i_{\text{attn}}$ means position i attends strongly to uncertain positions — a good candidate for remasking.

Ensemble uncertainty: Run the model forward twice with different dropout masks (or token dropout) and compare the two distributions at position i:

$$\widehat{I}^i_{\text{ens}} = \text{JSD}(p^{(1)}_\theta(x^i \mid z_t),\, p^{(2)}_\theta(x^i \mid z_t))$$

High Jensen-Shannon divergence indicates epistemic uncertainty (the model is inconsistent about this position). Cost: 2 forward passes per step.

9.7.3 Research Direction

Compare $\widehat{I}^i$, $\widehat{I}^i_{\text{MI}}$, $\widehat{I}^i_{\text{attn}}$, and $\widehat{I}^i_{\text{ens}}$ as remasking signals: 1. Calibration analysis: which signal best predicts actual prediction error? 2. Quality at matched compute: which signal achieves better MAUVE/gen_ppl per forward pass? 3. Regime analysis: which signal is most valuable early vs. late in generation?

9.8 R8: Statistical Significance and Evaluation Standards

9.8.1 The Problem

Every empirical result in this thesis and in the literature is a point estimate. At N=100 generated samples, MAUVE estimates carry substantial variance. Some "findings" may be noise: - MDLM gen_ppl worsening T=512→1000 (49.0→52.3): is this real? - remdm-loop T=512 MAUVE dip (0.614→0.532 before 0.684): is the dip real?

9.8.2 Formal Proposal: Bootstrap CI Framework

For MAUVE at N=100 samples and B=1000 bootstrap resamples:

$$[\text{MAUVE}^{(2.5\%)}, \text{MAUVE}^{(97.5\%)}] = \text{bootstrap CI at } 95\%$$

For gen_ppl, use a paired Wilcoxon signed-rank test between strategies (non-parametric, no normality assumption) to test whether the median per-sequence cross-entropy differs.

Minimum sample size. Define $N^*(\delta) = \min\{N : \text{CI width} < \delta\}$ for target precision $\delta = 0.05$ (5% of the MAUVE scale). Extrapolate from our N=100 data using the $O(1/\sqrt{N})$ scaling of standard errors.

Expected finding. Based on the variance in our results (MAUVE ranging 0.17–0.74 across strategies and step counts), N=100 likely gives 95% CIs of width ±0.10. The key differences (remdm-loop 0.684 vs. remdm-conf 0.325 at T=1000, Δ=0.359) are almost certainly significant. The small anomalies (gen_ppl 49.0→52.3) may not be.

Implementation. Can be run immediately on existing generated sequences in `results/*/generated_sequences.` Write `scripts/bootstrap_ci.py` to produce a table of CIs for the thesis. This requires no new HPC runs.

9.9 Summary Table: Research Directions

Direction	Training needed	New HPC runs	Theoretical	Empirical	Difficulty
R1: EB + Remasking	No	Yes	E_f interaction bound	Sweep over (ε, τ)	Medium
R2: Temperature calibration	No	Yes	Calibration analysis	Step sweep + temperature	Low
R3: Schedule co-design	Yes	Yes	Optimal schedule derivation	Train + inference	High
R4: Block-level reschedul.	No	Yes	Jensen's inequality bound	Block EB on LLaDA-8B	Medium
R5: Multi-pass convergence	No	Yes	Geometric bound on passes	K-pass sweep	Medium
R6: Conditional generation	No	Yes	Conditional I^i_{cond}	LAMBADA/HellaSwag	Medium
R7: Better info proxy	No	Yes	Calibration theory	Proxy comparison	Medium
R8: Statistical testing	No	No	Bootstrap theory	CIs from existing data	**Low**

For the thesis: R8 (statistical testing) should be done first — it validates or refutes existing results. R2 (temperature calibration) is the most tractable novel contribution: training-free, one extra hyperparameter, directly motivated by the remdm-conf collapse finding. R1 (EB + remasking) is the cleanest theoretical extension of the EB-Sampler paper and fills the most important gap in the literature.

9.10 The Rescheduling Problem: A Unified View

To close this part, here is the rescheduling problem stated as a single optimisation:

$$\min_{\pi \in \Pi} \mathrm{KL}(\pi_{\text{data}} \| p_\pi) \quad \text{subject to} \quad \mathrm{NFE}(\pi) \leq C$$

where: - Π is the space of all rescheduling policies (which tokens to unmask/remask, in what order, with what probabilities); - p_π is the distribution of sequences produced by policy π; - $\mathrm{NFE}(\pi)$ is the expected number of forward evaluations (compute budget); - C is the compute constraint.

Known solutions: - π = MDLM: uniform random unmasking, no remasking. $E_{\text{fact}} = O(1/T)$. - π = EB-Sampler: entropy-budget unmasking, no remasking. $E_{\text{fact}} = O(\varepsilon \cdot \max_i \widehat{I}^i)$ — optimal for unmasking-only policies. - π = thesis: EB-Sampler + confidence-guided remasking. $E_{\text{fact}} < E_{\text{EB}}$ when $\Sigma^2 > 0$. - π = RemeDi-RL: learned policy via RL. Empirically best but requires training.

Open: The globally optimal policy in Π (minimising KL subject to NFE budget) is unknown. The thesis derives the optimal policy within the class of *threshold* strategies. The globally optimal policy is presumably learned (RemeDi direction) but its form is not characterised theoretically.

9.11 Appendix A: Statistical and Mathematical Foundations

This appendix is a self-contained reference for the mathematical background required to understand discrete diffusion models. It covers probability theory, information theory, variational inference, Markov chains, and the specific structures that arise in masked diffusion. Each section includes definitions, key results, and their direct connections to the models reviewed in the main document.

9.11.1 B.1 Probability Theory Essentials

9.11.1.1 B.1.1 Random Variables and Distributions

A **random variable** X is a measurable function from a probability space $(\Omega, \mathcal{F}, P)$ to a measurable space. For discrete X taking values in a countable set $\mathcal{X}$:

- **Probability mass function (PMF):** $p(x) = P(X = x) \geq 0$, $\sum_{x \in \mathcal{X}} p(x) = 1$
- **Support:** $\text{supp}(p) = \{x : p(x) > 0\}$

Categorical distribution $\text{Cat}(\pi)$ over $\mathcal{X} = \{1, \ldots, V\}$:

$$P(X = v) = \pi_v, \quad \pi \in \Delta_{V-1}$$

where $\Delta_{V-1} = \{\pi \in \mathbb{R}^V : \pi_v \geq 0,\ \sum_v \pi_v = 1\}$ is the $(V-1)$-dimensional probability simplex.

One-hot encoding: $\mathbf{e}_v \in \{0, 1\}^V$ has $(\mathbf{e}_v)_i = \mathbf{1}[i = v]$. Then $X \sim \text{Cat}(\pi)$ can be written $P(X = v) = \pi^\top \mathbf{e}_v$.

9.11.1.2 B.1.2 Conditional Probability and Bayes' Theorem

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Bayes' theorem:

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}$$

where $P(B) = \sum_a P(B \mid A = a)\, P(A = a)$ is the normalising constant (marginal likelihood).

In the diffusion context: $P(\mathbf{x}_0 \mid \mathbf{x}_t)$ is the posterior over clean sequences given a noisy (masked) observation, computed via Bayes from the forward kernel $P(\mathbf{x}_t \mid \mathbf{x}_0)$ and the prior $P(\mathbf{x}_0)$.

9.11.1.3 B.1.3 Independence and Conditional Independence

X and Y are **independent** ($X \perp Y$) iff $P(X, Y) = P(X)\,P(Y)$ for all values.

X and Y are **conditionally independent given** Z ($X \perp Y \mid Z$) iff $P(X, Y \mid Z) = P(X \mid Z)\,P(Y \mid Z)$.

Why this matters for masked diffusion: The reverse model $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ is typically factored as $\prod_i p_\theta(x_{t-1}^i \mid \mathbf{x}_t)$, i.e., positions are treated as conditionally independent given $\mathbf{x}_t$. The true posterior $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ does **not** factorise this way — tokens are jointly dependent — and this discrepancy is the **factorisation error** E_{fact}.

9.11.1.4 B.1.4 Expectation and Variance

$$\mathbb{E}[X] = \sum_x x\,p(x), \qquad \mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Law of total expectation: $\mathbb{E}[X] = \mathbb{E}_Y[\mathbb{E}[X \mid Y]]$

Law of total variance: $\mathrm{Var}(X) = \mathbb{E}_Y[\mathrm{Var}(X \mid Y)] + \mathrm{Var}_Y(\mathbb{E}[X \mid Y])$

9.11.1.5 B.1.5 Jensen's Inequality

For a **convex** function f (i.e., $f''(x) \geq 0$):

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

For a **concave** function ($f''(x) \leq 0$), the inequality reverses: $f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$.

Critical application: $-\log$ is convex, so $-\log(\mathbb{E}[Z]) \leq \mathbb{E}[-\log Z]$, which gives the ELBO $\leq$ log-marginal inequality (see Section B.3).

9.11.2 B.2 Information Theory

All logarithms are natural (ln) unless stated; entropy is in **nats**. To convert to bits, divide by $\ln 2$.

9.11.2.1 B.2.1 Entropy

The **Shannon entropy** of a discrete random variable $X \sim p$:

$$H(X) = -\sum_{x \in \mathcal{X}} p(x) \ln p(x)$$

(convention: $0 \ln 0 = 0$).

Key properties:

Property	Statement	Proof sketch
Non-negativity	$H(X) \geq 0$	$p(x) \leq 1 \Rightarrow -\ln p(x) \geq 0$
Determinism	$H(X) = 0 \iff X$ is deterministic	Only one $p(x) > 0$
Maximum	$H(X) \leq \ln \lvert\mathcal{X}\rvert$	Lagrange multipliers; equality iff uniform

Property	Statement	Proof sketch
Concavity	H is concave in p	Hessian is negative semidefinite

Interpretation: $H(X)$ measures the average uncertainty about X before observing it, or equivalently the average code length (in nats) under an optimal code.

Character perplexity: $e^{H(X)}$ (or $2^{H(X)}$ in bits) is the *effective alphabet size* in the sense that a uniform distribution over $e^{H(X)}$ symbols has the same entropy. For a token entropy of $H = 5.5$ nats: character perplexity $= e^{5.5} \approx 245$. An entropy change ΔH corresponds to a character perplexity ratio of $e^{\Delta H}$.

9.11.2.2 B.2.2 Joint and Conditional Entropy

$$H(X, Y) = -\sum_{x,y} p(x, y) \ln p(x, y)$$

$$H(X \mid Y) = -\sum_{x,y} p(x, y) \ln p(x \mid y) = H(X, Y) - H(Y)$$

Conditioning reduces entropy: $H(X \mid Y) \leq H(X)$, with equality iff $X \perp Y$.

Chain rule: $H(X_1, X_2, \ldots, X_n) = \sum_{i=1}^{n} H(X_i \mid X_1, \ldots, X_{i-1})$

This is the factorisation of the joint distribution into a product of conditionals: $p(x_1, \ldots, x_n) = \prod_i p(x_i \mid x_1, \ldots, x_{i-1})$.

For sequences: A language model p_θ assigns probability $p_\theta(\mathbf{x}) = \prod_{i=1}^{L} p_\theta(x^i \mid x^1, \ldots, x^{i-1})$, computing each conditional autoregressively. Masked diffusion models compute all conditionals in parallel (one forward pass), which is efficient but introduces the factorisation error when tokens are decoded jointly.

9.11.2.3 B.2.3 KL Divergence

The **Kullback-Leibler divergence** from Q to P (read: "P from Q" or "KL of P with respect to Q"):

$$D_{\mathrm{KL}}(P \| Q) = \sum_{x} p(x) \ln \frac{p(x)}{q(x)}$$

Key properties:

- **Non-negativity (Gibbs' inequality):** $D_{\mathrm{KL}}(P \| Q) \geq 0$, with equality iff $P = Q$ a.e. *Proof:* $-D_{\mathrm{KL}} = \mathbb{E}_P[\ln(q/p)] \leq \ln \mathbb{E}_P[q/p] = \ln 1 = 0$ (Jensen, ln concave).
- **Asymmetry:** $D_{\mathrm{KL}}(P \| Q) \neq D_{\mathrm{KL}}(Q \| P)$ in general.
- **Mode-seeking vs mode-covering:**
  - Minimising $D_{\mathrm{KL}}(q \| p)$ w.r.t. q: q concentrates on modes of p (mode-seeking, zero-forcing). If $p(x) > 0$ but $q(x) = 0$, the term $q(x) \ln(q(x)/p(x)) = 0$ — no penalty for missing mass.

- Minimising $D_{\mathrm{KL}}(p\|q)$ w.r.t. q: q must cover all mass of p (mode-covering, mean-seeking). If $p(x) > 0$ but $q(x) \approx 0$, the term $p(x)\ln(p(x)/q(x)) \to +\infty$ — severe penalty.

This asymmetry is empirically observable in our results: remdm-conf at T=1000 achieves low gen_ppl (model fits GPT-2 distribution well by mode-seeking) but low MAUVE (fails to cover the full OWT distribution). The mode-seeking direction minimises one KL; covering diversity requires minimising the other.

9.11.2.4 B.2.4 Cross-Entropy

$$H(P,Q) = -\sum_x p(x)\ln q(x) = H(P) + D_{\mathrm{KL}}(P\|Q)$$

Interpretation: The average code length when data is distributed as P but coded under Q. Always $H(P,Q) \geq H(P)$, with equality iff $P = Q$.

Gen-ppl decomposition (critical for thesis):

$$\text{gen-ppl} = \exp H(p_\theta, q_{\text{GPT-2}}) = \exp[H(p_\theta) + D_{\mathrm{KL}}(p_\theta\|q_{\text{GPT-2}})]$$

A low gen-ppl can arise from: (a) low entropy $H(p_\theta)$ (mode collapse), (b) low KL (distribution aligned with GPT-2), or (c) both. It is **not** a pure measure of distributional fit — a degenerate model that generates a single, GPT-2-likely sentence would achieve gen-ppl ≈ 1 while being useless.

9.11.2.5 B.2.5 Mutual Information

$$I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X) = D_{\mathrm{KL}}(P_{XY}\|P_X P_Y)$$

- $I(X;Y) \geq 0$, equal to zero iff $X \perp Y$.
- $I(X;Y) = I(Y;X)$ (symmetric, unlike KL).
- $I(X;Y) = H(X) + H(Y) - H(X,Y)$ (visualised via the entropy Venn diagram).

Data processing inequality (DPI): If $X \to Y \to Z$ is a Markov chain (i.e., $Z \perp X \mid Y$), then $I(X;Z) \leq I(X;Y)$.

DPI in diffusion: The chain $\mathbf{x}_0 \to \mathbf{x}_t \to \mathbf{x}_T$ means $I(\mathbf{x}_0;\mathbf{x}_T) \leq I(\mathbf{x}_0;\mathbf{x}_t)$ for $t \leq T$. The forward process monotonically destroys information about $\mathbf{x}_0$. As $\alpha_T \to 0$ (all tokens masked), $I(\mathbf{x}_0;\mathbf{x}_T) \to 0$.

9.11.2.6 B.2.6 Information Profile (connection to thesis)

The **information profile** of position i in sequence $\mathbf{x}$ is:

$$I^i(\mathbf{x}) = H(x^i \mid \mathbf{x}^{\ i})$$

This is the conditional entropy of position i given all other positions. It measures how much uncertainty about x^i remains after observing all other tokens.

- $I^i(\mathbf{x}) = 0$: position i is fully determined by context (e.g., a function word after a fixed phrase).
- $I^i(\mathbf{x})$ large: position i is highly variable given context (e.g., the first content word of a sentence).

Lavenant & Zanella (2024) prove that the optimal unmasking order is increasing in $I^i(\mathbf{x})$: unmask positions with **lowest** I^i first (most predictable given context). This minimises the factorisation error at each step — you never commit a token that is highly uncertain given its neighbours.

9.11.3 B.3 Variational Inference and the ELBO

9.11.3.1 B.3.1 Latent Variable Models A **latent variable model** defines a joint distribution $p_\theta(\mathbf{x}, \mathbf{z})$ over observed variables $\mathbf{x}$ and latent variables $\mathbf{z}$:

$$p_\theta(\mathbf{x}, \mathbf{z}) = p_\theta(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z})$$

The **marginal likelihood** (evidence):

$$p_\theta(\mathbf{x}) = \sum_{\mathbf{z}} p_\theta(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z})$$

is typically **intractable** (sum over all latent configurations). Maximum likelihood training requires maximising $\log p_\theta(\mathbf{x})$, which requires computing this sum.

In discrete diffusion: $\mathbf{x}$ is the clean sequence $\mathbf{x}_0$ and $\mathbf{z} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$ is the noisy trajectory. $p(\mathbf{z})$ is the forward process (fixed), and $p_\theta(\mathbf{x}_0 \mid \mathbf{z})$ is the reverse model (learned).

9.11.3.2 B.3.2 The Evidence Lower Bound (ELBO) Introduce a variational distribution $q_\phi(\mathbf{z} \mid \mathbf{x})$ to approximate the intractable posterior $p_\theta(\mathbf{z} \mid \mathbf{x})$. Then:

$$\log p_\theta(\mathbf{x}) = \log \sum_{\mathbf{z}} q_\phi(\mathbf{z} \mid \mathbf{x}) \frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z} \mid \mathbf{x})}$$

Applying Jensen's inequality (log is concave, so $\log \mathbb{E}[Z] \geq \mathbb{E}[\log Z]$):

$$\log p_\theta(\mathbf{x}) \geq \mathbb{E}_{q_\phi}\left[\log \frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z} \mid \mathbf{x})}\right] =: \mathcal{L}(\theta, \phi; \mathbf{x}) \quad \text{(ELBO)}$$

Two equivalent forms:

$$\mathcal{L} = \underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x} \mid \mathbf{z})]}_{\text{reconstruction}} - \underbrace{D_{\mathrm{KL}}(q_\phi(\mathbf{z} \mid \mathbf{x}) \| p(\mathbf{z}))}_{\text{regularisation}}$$

$$\log p_\theta(\mathbf{x}) = \mathcal{L}(\theta, \phi; \mathbf{x}) + \underbrace{D_{\mathrm{KL}}(q_\phi(\mathbf{z} \mid \mathbf{x}) \| p_\theta(\mathbf{z} \mid \mathbf{x}))}_{\geq 0}$$

The second form shows the ELBO is a lower bound on the log-evidence, with the gap equal to the KL between the variational posterior and the true posterior. The ELBO is tight ($= \log p_\theta(\mathbf{x})$) when $q_\phi = p_\theta(\cdot \mid \mathbf{x})$.

9.11.3.3 B.3.3 ELBO for Diffusion Models

In discrete diffusion (MDLM, D3PM), the ELBO expands as:

$$\mathcal{L} = \underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}[\log p_\theta(\mathbf{x}_0 \mid \mathbf{x}_1)]}_{\text{reconstruction at } t=1} - \underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T \mid \mathbf{x}_0)\|p(\mathbf{x}_T))}_{\text{prior matching}} - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[D_{\mathrm{KL}}(q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)\|p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t))]}_{\text{denoising terms}}$$

Key insight: Each denoising term measures how well $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ matches the *true* Bayesian reverse $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$. For the absorbing process, this posterior has an analytic closed form (see Section B.6), which is why MDLM training is tractable: the model just needs to learn to predict $\mathbf{x}_0$ from $\mathbf{x}_t$, and the posterior is computed analytically.

9.11.3.4 B.3.4 Mean-Field Approximation

The **mean-field** variational family assumes full factorisation: $q(\mathbf{z}) = \prod_i q_i(z_i)$.

The optimal factor satisfies: $\ln q_j^*(z_j) = \mathbb{E}_{-j}[\ln p(\mathbf{x}, \mathbf{z})] + \text{const}$

Connection to factorisation error: The reverse model $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \prod_i p_\theta(x_{t-1}^i \mid \mathbf{x}_t)$ is a mean-field approximation to the joint reverse posterior. The factorisation error is exactly the KL penalty from using this mean-field approximation instead of the true joint posterior.

9.11.4 B.4 Markov Chains

9.11.4.1 B.4.1 Definition and Markov Property

A sequence of random variables $(X_0, X_1, X_2, \ldots)$ is a **Markov chain** if:

$$P(X_t = x_t \mid X_0 = x_0, \ldots, X_{t-1} = x_{t-1}) = P(X_t = x_t \mid X_{t-1} = x_{t-1})$$

The future state depends only on the present state, not the past. This is called the **Markov property** (memorylessness).

A chain is **time-homogeneous** if the transition kernel $T(x, y) = P(X_t = y \mid X_{t-1} = x)$ does not depend on t.

9.11.4.2 B.4.2 Transition Matrix and Chapman-Kolmogorov

For a finite state space $\mathcal{S}$, the transition kernel is a matrix $\mathbf{T}$ with: - $T_{xy} = P(X_t = y \mid X_{t-1} = x) \geq 0$ - $\sum_y T_{xy} = 1$ (rows sum to one — $\mathbf{T}$ is a **stochastic matrix**)

k-step transition: $P(X_{t+k} = y \mid X_t = x) = (\mathbf{T}^k)_{xy}$

Chapman-Kolmogorov: $\mathbf{T}^{t+s} = \mathbf{T}^t \cdot \mathbf{T}^s$

In the diffusion context, the t-step marginal of the forward process $q(\mathbf{x}_t \mid \mathbf{x}_0)$ is computed by applying $\mathbf{T}^t$ to the one-hot encoding $\mathbf{e}_{x_0}$.

9.11.4.3 B.4.3 Stationary Distribution and Detailed Balance

A distribution π is **stationary** for $\mathbf{T}$ if $\pi = \pi\mathbf{T}$, i.e., $\pi(y) = \sum_x \pi(x)T(x, y)$.

A chain satisfies **detailed balance** (is **reversible**) if:

$$\pi(x)\, T(x, y) = \pi(y)\, T(y, x) \quad \forall x, y$$

Detailed balance implies stationarity (but not vice versa). Reversibility means the chain looks statistically identical run forwards or backwards in time — a property that diffusion models exploit: the forward (noising) and reverse (denoising) processes share the same stationary distribution.

Absorbing state: State a is absorbing if $T(a, a) = 1$ (and $T(a, y) = 0$ for $y \neq a$). For the masking process: `[MASK]` is absorbing. Once a token is masked, it stays masked under the forward process. The reverse process p_θ must learn to "unmask" it.

9.11.4.4 B.4.4 Continuous-Time Markov Chains (CTMC)

A **CTMC** is a Markov chain with continuous time index $t \in [0, \infty)$. It is characterised by a **rate matrix** (generator) $\mathbf{Q}$: - $Q_{ij} \geq 0$ for $i \neq j$ (rates of transitioning from i to j) - $Q_{ii} = -\sum_{j \neq i} Q_{ij}$ (rows sum to zero)

Transition matrix: $\mathbf{P}(t) = e^{\mathbf{Q}t} = \sum_{k=0}^{\infty} \frac{(Qt)^k}{k!}$

Kolmogorov forward equation: $\frac{d}{dt}\mathbf{P}(t) = \mathbf{P}(t)\,\mathbf{Q}$

Connection to discrete diffusion: SEDD and DFM (Campbell et al.) formulate the forward process as a CTMC with $Q_{x,\texttt{MASK}} = \sigma(t)$ for all $x \neq \texttt{MASK}$, recovering the absorbing discrete diffusion process in continuous time. The reverse CTMC has rate matrix characterised by the score function of the marginal distribution.

9.11.4.5 B.4.5 The Backward Equation and Reverse Process

For a forward CTMC with generator $\mathbf{Q}$ and stationary distribution π, the **time-reversal** is also a CTMC with rates:

$$\tilde{Q}_{ij}(t) = \frac{\pi_j(t)}{\pi_i(t)} Q_{ji}$$

This is the continuous-time analogue of the reversal used in diffusion models. The key insight: **learning the reverse process is equivalent to learning the ratios** π_j/π_i, which connects to score-based methods.

9.11.5 B.5 Score Functions and Denoising

(Primarily for continuous-space context; included for completeness and for understanding SEDD.)

9.11.5.1 B.5.1 Score Function

The **score function** of a distribution $p(\mathbf{x})$ over $\mathbf{x} \in \mathbb{R}^d$:

$$s(\mathbf{x}) = \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

Points in the direction of increasing log-density — toward regions of higher probability mass.

Example: For $p(\mathbf{x}) = \mathcal{N}(\mu, \Sigma)$: $s(\mathbf{x}) = -\Sigma^{-1}(\mathbf{x} - \mu)$

The score of a Gaussian points back toward the mean — it is a linear "restoring force."

9.11.5.2 B.5.2 Fisher Divergence and Score Matching

The **Fisher divergence** between distributions p and q:

$$D_F(p\|q) = \mathbb{E}_p\left[\|s_p(\mathbf{x}) - s_q(\mathbf{x})\|^2\right]$$

Score matching (Hyvärinen 2005) minimises $D_F(p_{\text{data}}\|p_\theta)$ without requiring p_{data} to be tractable, using integration by parts.

9.11.5.3 B.5.3 Denoising Score Matching

Denoising score matching (Vincent 2011) adds Gaussian noise $\tilde{\mathbf{x}} = \mathbf{x} + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$, and trains a denoiser $s_\theta(\tilde{\mathbf{x}})$ to match $\nabla_{\tilde{\mathbf{x}}} \log p(\tilde{\mathbf{x}} \mid \mathbf{x})$:

$$\min_\theta \mathbb{E}_{\mathbf{x}\sim p} \mathbb{E}_\epsilon \left[\left\| s_\theta(\tilde{\mathbf{x}}) + \frac{\epsilon}{\sigma^2} \right\|^2\right]$$

This avoids computing intractable score of p_{data} directly.

Tweedie's formula: For $\tilde{\mathbf{x}} = \mathbf{x}_0 + \sigma\epsilon$:

$$\mathbb{E}[\mathbf{x}_0 \mid \tilde{\mathbf{x}}] = \tilde{\mathbf{x}} + \sigma^2 \nabla_{\tilde{\mathbf{x}}} \log p(\tilde{\mathbf{x}})$$

The posterior mean of the clean signal equals the noisy signal plus a score-weighted correction. This underlies the DDPM $\mathbf{x}_0$-prediction parametrisation.

9.11.5.4 B.5.4 Discrete Score Function (SEDD)

For a discrete distribution $p(\mathbf{x})$ over $\mathcal{X}^L$, there is no gradient. The **concrete score** (Lou et al., 2023) at position i and value v:

$$s_\theta(\mathbf{x})^i_v = \frac{p_\theta([\mathbf{x}^{\,i}, v])}{p_\theta(\mathbf{x})}$$

This ratio measures how the probability changes when position i is changed to v. The model learns these ratios directly, and they are used to define the reverse CTMC. SEDD shows this leads to tractable training via a **score-entropy** loss.

9.11.6 B.6 The Absorbing (Masking) Forward Process

This section details the specific Markov chain used in MDLM, D3PM (absorbing), ReMDM, LLaDA.

9.11.6.1 B.6.1 Forward Kernel

Let $\mathcal{V} = \{1, \dots, V\} \cup \{\mathtt{M}\}$ where $\mathtt{M}$ is the mask token. The **one-step forward kernel** masks each token independently:

$$q(x_t^i \mid x_{t-1}^i) = (1 - \beta_t)\, \mathbf{e}_{x_{t-1}^i} + \beta_t\, \mathbf{e}_{\mathtt{M}}$$

where $\beta_t \in [0, 1]$ is the masking probability at step t.

t-step marginal (by composing the one-step kernels):

$$q(x_t^i \mid x_0^i) = \alpha_t\, \mathbf{e}_{x_0^i} + (1 - \alpha_t)\, \mathbf{e}_{\mathtt{M}}$$

where $\alpha_t = \prod_{s=1}^{t}(1 - \beta_s)$ is the **survival probability** — the probability a token remains unmasked at step t. $\alpha_0 = 1$ (fully clean), $\alpha_T \approx 0$ (fully masked).

Simplified: At time t, position i is: - Unmasked (equals x_0^i) with probability α_t - Masked ($= \mathtt{M}$) with probability $1 - \alpha_t$

The positions are masked **independently** — the forward process factorises over positions.

9.11.6.2 B.6.2 Posterior (Reverse Bayesian Update)

The posterior $q(x_{t-1}^i \mid x_t^i, x_0^i)$ is computed by Bayes:

Case 1: $x_t^i \neq \mathtt{M}$ (token is already unmasked):

$$q(x_{t-1}^i \mid x_t^i \neq \mathtt{M}, x_0^i) = \delta(x_{t-1}^i = x_t^i)$$

If a token is visible at time t, it was also visible at $t - 1$ (absorbing state property). The reverse step is deterministic: keep the token.

Case 2: $x_t^i = \mathtt{M}$ (token is masked):

$$q(x_{t-1}^i \mid x_t^i = \mathtt{M}, x_0^i) = \frac{\alpha_{t-1} - \alpha_t}{1 - \alpha_t}\, \mathbf{e}_{x_0^i} + \frac{1 - \alpha_{t-1}}{1 - \alpha_t}\, \mathbf{e}_{\mathtt{M}}$$

Derivation: By Bayes, $q(x_{t-1}^i \mid \mathtt{M}, x_0^i) \propto q(\mathtt{M} \mid x_{t-1}^i)\, q(x_{t-1}^i \mid x_0^i)$. Two cases for x_{t-1}^i: - $x_{t-1}^i = x_0^i$ (unmasked): $q(\mathtt{M} \mid x_0^i) \cdot \alpha_{t-1} = (1 - \alpha_{t-1}) \cdot \alpha_{t-1}$… wait, more carefully: $q(\mathtt{M} \mid x_{t-1}^i = x_0^i) = 1 - \alpha_t/\alpha_{t-1}$ (one-step mask probability given unmasked at $t - 1$) $q(x_{t-1}^i = x_0^i \mid x_0^i) = \alpha_{t-1}$ Weight: $\propto (1 - \alpha_t/\alpha_{t-1}) \cdot \alpha_{t-1} = \alpha_{t-1} - \alpha_t$ - $x_{t-1}^i = \mathtt{M}$ (masked): $q(\mathtt{M} \mid \mathtt{M}) = 1$ (absorbing), $q(\mathtt{M} \mid x_0^i) = 1 - \alpha_{t-1}$ Weight: $\propto 1 \cdot (1 - \alpha_{t-1}) = 1 - \alpha_{t-1}$

Normalising by $1 - \alpha_t = (\alpha_{t-1} - \alpha_t) + (1 - \alpha_{t-1})$ gives the formula above.

Key insight: Given that a token is masked at time t, the posterior says: unmask it to x_0^i with probability $(\alpha_{t-1} - \alpha_t)/(1 - \alpha_t)$, or keep it masked with the complementary probability. The model must learn to predict x_0^i from $\mathbf{x}_t$ to implement this reverse step.

9.11.6.3 B.6.3 Masked Cross-Entropy Training Loss

The ELBO denoising term at step t:

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[D_{\mathrm{KL}}(q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t))]$$

reduces (using the closed-form posterior) to a **masked cross-entropy**:

$$\mathcal{L}_t \propto -\sum_{i:\, x_t^i = \mathtt{M}} \mathbb{E}[\ln p_\theta(x_0^i \mid \mathbf{x}_t)]$$

Only masked positions contribute to the loss. The model is trained to predict the original token x_0^i at each masked position given the partially-masked sequence $\mathbf{x}_t$.

This is exactly masked language modelling (BERT-style), but with a principled derivation: it is the ELBO of the absorbing diffusion model. The difference from BERT: the masking schedule α_t is explicit, and the loss is summed over all t during training.

9.11.6.4 B.6.4 Noise Schedule

The noise schedule $\{\alpha_t\}_{t=0}^T$ controls how quickly tokens are masked.

Linear schedule: $\alpha_t = 1 - t/T$ — uniform masking across steps.

Cosine schedule (MDLM): $\alpha_t = \cos^2(\frac{\pi t}{2T})$ — slow start and end, faster masking in the middle. Inspired by the continuous-time flow matching literature.

Effect on training: The schedule determines what fraction of tokens are masked at each training step. A cosine schedule ensures the model trains on a roughly uniform distribution over masking rates, unlike linear which concentrates training at intermediate rates.

9.11.7 B.7 Factorisation Error: Derivation and Bounds

9.11.7.1 B.7.1 Where the Error Comes from

The true joint reverse process:

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \sum_{\mathbf{x}_0} q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)\, q(\mathbf{x}_0 \mid \mathbf{x}_t)$$

This is a mixture over all possible clean sequences — intractable, and **does not factorise** over positions (the joint distribution of all unmasked tokens is correlated through $\mathbf{x}_0$).

The model approximates:

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \prod_i p_\theta(x_{t-1}^i \mid \mathbf{x}_t)$$

The error per step is:

$$\varepsilon_t = D_{\mathrm{KL}}\left(\prod_i q(x_{t-1}^i \mid \mathbf{x}_t, \mathbf{x}_0) \middle\| q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)\right)$$

Summing over steps: $E_{\mathrm{fact}} = \sum_{t=1}^T \varepsilon_t$.

9.11.7.2 B.7.2 Scaling with T

Claim: For smooth models, $E_{\text{fact}} = O(\Sigma^2/T)$ where Σ^2 measures the covariance between positions in the posteriors.

Intuition: At each step, the fraction of newly unmasked tokens is $\Delta\alpha_t = \alpha_{t-1} - \alpha_t \approx 1/T$. The cross-correlations between unmasked tokens scale as $O(\Delta\alpha_t^2) = O(1/T^2)$. Summing over T steps: $E_{\text{fact}} \sim T \cdot O(1/T^2) = O(1/T)$.

Consequence: More steps → lower factorisation error → better generation quality. This motivates using large T (e.g., T=1000 vs T=128). The MAUVE improvement from T=128 to T=1000 in our results is at least partially explained by E_{fact} reduction.

9.11.7.3 B.7.3 Remasking as Error Correction

Remasking addresses E_{fact} at inference time without retraining. If a token x^i was committed (unmasked) based on insufficient context — because correlated tokens x^j, x^k were still masked — remasking it to `M` allows the model to reconsider it after those context tokens are revealed.

Formal effect: Let $E_{\text{fact}}^{\text{base}}$ be the factorisation error without remasking. With remasking, committed tokens can be revised, reducing the effective Σ^2 at each step. The theoretical gain depends on whether the proxy confidence signal aligns with the information profile $I^i(\mathbf{x})$.

9.11.8 B.8 Evaluation Metrics: Formal Definitions

9.11.8.1 B.8.1 Perplexity and Bits-Per-Byte

Perplexity of a language model p_θ on text data $\mathcal{D}$:

$$\text{PPL}(p_\theta, \mathcal{D}) = \exp\left(\frac{1}{N}\sum_{n=1}^{N} -\log p_\theta(\mathbf{x}_n)\right) = \exp\left(H(p_{\text{data}}, p_\theta)\right)$$

where $H(p_{\text{data}}, p_\theta)$ is the cross-entropy of the data distribution under the model.

Bits-per-byte (BPB): $\text{BPB} = H(p_{\text{data}}, p_\theta)/\ln(2)/C$ where C is the average number of bytes per token (typically ≈ 4 for GPT-2 tokenizer).

Gen-ppl (used in this thesis): evaluates the *generated* distribution p_θ under an external evaluator $q_{\text{GPT-2}}$:

$$\text{gen-ppl} = \exp H(p_\theta, q_{\text{GPT-2}}) = \exp\left[H(p_\theta) + D_{\text{KL}}(p_\theta \| q_{\text{GPT-2}})\right]$$

This is estimated by: generate N sequences from p_θ, compute their average negative log-likelihood under GPT-2. Lower gen-ppl means generated text is more likely under GPT-2 (closer to fluent English), **but** it conflates fluency (low KL) with low diversity (low entropy). A model that generates one GPT-2-likely sentence repeatedly achieves gen-ppl ≈ 1 while being degenerate.

9.11.8.2 B.8.2 MAUVE

MAUVE (Pillutla et al., 2021) measures the divergence between the model distribution P (generated samples) and the reference distribution Q (human text) via the **divergence frontier**.

For $\lambda \in [0, 1]$, define the mixture:

$$M_\lambda = (1-\lambda)\, P + \lambda\, Q$$

The **divergence frontier** is the curve:

$$\mathcal{F}(P,Q) = \{(\mathrm{KL}(M_\lambda \| P),\ \mathrm{KL}(M_\lambda \| Q)) : \lambda \in [0,1]\}$$

This traces a curve in $\mathbb{R}^2_{\geq 0}$. When $P = Q$, $M_\lambda = P = Q$ for all λ, and both KLs are zero — the curve collapses to the origin.

MAUVE score:

$$\mathrm{MAUVE}(P,Q) = \exp(-c \cdot \mathrm{Area}(\mathcal{F}(P,Q)))$$

where $c > 0$ is a scaling constant. Lower area = higher MAUVE = distributions more similar. MAUVE $\in (0,1]$, with MAUVE $= 1$ when $P = Q$.

Geometric interpretation of the frontier: - The x-axis KL = $\mathrm{KL}(M_\lambda \| P)$ measures how far the mixture deviates from P (model). High λ (mixture close to Q) $\to$ large x. - The y-axis KL = $\mathrm{KL}(M_\lambda \| Q)$ measures how far the mixture deviates from Q (reference). Low λ (mixture close to P) $\to$ large y. - If P has low recall (misses modes of Q): large y-axis values (high divergence from Q). - If P has low precision (generates OOD text): large x-axis values (high divergence from P).

Practical computation: P and Q are not available analytically; they are represented as sets of feature vectors (GPT-2 embeddings of generated / reference sentences). The frontier is approximated by a quantisation of the feature space. This introduces sampling variance — estimates are unreliable for $N < 500$ samples (Pillutla et al.).

9.11.8.3 B.8.3 Entropy as a Diversity Proxy

Token-level entropy of the generated distribution:

$$H_{\text{gen}} = \frac{1}{L}\sum_{i=1}^{L} H(x^i) = -\frac{1}{L}\sum_{i=1}^{L}\sum_{v} p_\theta(x^i = v)\,\ln p_\theta(x^i = v)$$

In practice, estimated by the empirical marginal over N generated samples.

Interpretation: High entropy = diverse vocabulary usage; low entropy = repetitive text. Entropy drop is a **signature of mode collapse**: the model is concentrating on a small subset of tokens. In our results, remdm-conf drops from $H = 5.499$ at T=128 to $H = 5.357$ at T=1000 — a decrease of 0.142 nats, corresponding to a character perplexity ratio of $e^{0.142} \approx 1.15$ (15% reduction in effective vocabulary).

Limitations: Entropy is a marginal statistic — it misses higher-order structure. A model that generates each token independently at the correct marginal frequency would have the correct entropy but zero inter-token correlation (wrong text). Entropy is necessary but not sufficient for quality assessment.

9.11.8.4 B.8.4 Bootstrap Confidence Intervals

Given N i.i.d. samples and a statistic $\hat{\theta}$ (e.g., MAUVE, gen-ppl), the **bootstrap** (Efron, 1979) estimates the sampling distribution of $\hat{\theta}$:

1. Draw B bootstrap samples: for each $b = 1, \ldots, B$, resample N observations with replacement from the original N samples.
2. Compute $\hat{\theta}^{(b)}$ for each bootstrap sample.
3. The bootstrap distribution $\{\hat{\theta}^{(b)}\}_{b=1}^{B}$ approximates the sampling distribution of $\hat{\theta}$.

Percentile CI: $[\hat{\theta}^{(\alpha/2)}, \hat{\theta}^{(1-\alpha/2)}]$ where the quantiles are from the bootstrap distribution. This is a 95% CI when $\alpha = 0.05$.

Why N=100 is marginal for MAUVE: MAUVE estimates the area of a frontier in feature space — it requires enough samples to cover the manifold. With $N = 100$, bootstrap CIs typically span ± 0.05–0.10 MAUVE units. A difference of $\Delta\text{MAUVE} = 0.09$ (e.g., remdm-conf vs remdm-loop at T=128: 0.440 vs 0.396) may not be significant. Bootstrap CIs are needed to determine which differences in the step-sweep table are reliably non-zero.

9.11.9 B.9 Softmax, Temperature Scaling, and Calibration

9.11.9.1 B.9.1 Softmax

The **softmax** function maps logits $\mathbf{z} \in \mathbb{R}^V$ to probabilities:

$$\text{softmax}(\mathbf{z})_v = \frac{e^{z_v}}{\sum_{v'=1}^{V} e^{z_{v'}}}$$

Temperature softmax with temperature $\tau > 0$:

$$\text{softmax}_\tau(\mathbf{z})_v = \frac{e^{z_v/\tau}}{\sum_{v'} e^{z_{v'}/\tau}}$$

- $\tau \to 0$: $\text{softmax}_\tau \to \arg\max$ (one-hot, deterministic greedy)
- $\tau = 1$: standard softmax
- $\tau \to \infty$: $\text{softmax}_\tau \to \text{Uniform}(V)$

Temperature annealing in remasking: At high masking rates (early generation), the model is uncertain — high temperature encourages exploration. As fewer tokens are masked, the model is more confident — lower temperature enables exploitation. The schedule $\tau(t) = 1 + (\tau_0 - 1)(t/T)$ achieves $\tau = \tau_0 > 1$ at $t = T$ (fully masked) and $\tau = 1$ at $t = 0$ (fully revealed).

9.11.9.2 B.9.2 Calibration

A model is **calibrated** if its confidence equals its empirical accuracy:

$$P(\hat{y} = y \mid p_\theta(\hat{y}) = p) = p \quad \forall p \in [0, 1]$$

Overconfidence: $p_\theta(\hat{y}) > P(\text{correct})$ — the model claims higher certainty than warranted. Common in large neural language models.

Temperature scaling is the simplest post-hoc calibration method: find τ^* on a held-out validation set that minimises the expected calibration error (ECE):

$$\tau^* = \arg\min_\tau \frac{1}{N} \sum_n \text{NLL}_\tau(y_n, p_\theta(\cdot \mid \mathbf{x}_n))$$

Connection to remdm-conf collapse: If p_θ is overconfident (too-small τ), the confidence signal is overconfident — it commits tokens prematurely. At T=1000 there are 1000 opportunities to make premature commitments that cannot be revisited, leading to diversity collapse. Entropy-based or temperature-calibrated confidence is a principled fix.

9.11.10 B.10 Worked Examples: Applying the Formulas

9.11.10.1 B.10.1 Computing KL Divergence for a Categorical Let $P = \mathrm{Cat}(0.7, 0.2, 0.1)$ and $Q = \mathrm{Cat}(0.5, 0.3, 0.2)$ over $V = 3$ tokens.

$$\begin{aligned}D_{\mathrm{KL}}(P\|Q) &= 0.7\ln\frac{0.7}{0.5} + 0.2\ln\frac{0.2}{0.3} + 0.1\ln\frac{0.1}{0.2} \\ &= 0.7 \times 0.336 + 0.2 \times (-0.405) + 0.1 \times (-0.693) \\ &= 0.235 - 0.081 - 0.069 = 0.085 \text{ nats}\end{aligned}$$

Note: the first term dominates because P has high mass on $v = 1$ but Q assigns less. The mode-seeking asymmetry: $D_{\mathrm{KL}}(Q\|P) = 0.5\ln(0.5/0.7) + ... \approx 0.076$ nats — different and slightly smaller (the reverse direction is less penalised for the mass shift).

9.11.10.2 B.10.2 ELBO Bound Tightness Suppose $\log p_\theta(\mathbf{x}) = -3.2$ nats. A variational approximation achieves $\mathcal{L} = -4.1$ nats. Then:

$$D_{\mathrm{KL}}(q\|p_\theta(\cdot \mid \mathbf{x})) = \log p_\theta(\mathbf{x}) - \mathcal{L} = -3.2 - (-4.1) = 0.9 \text{ nats}$$

The variational posterior is 0.9 nats from the true posterior — substantial room for improvement. In practice, a well-trained diffusion model drives this gap toward zero by the design of the forward process (the posterior is analytically tractable).

9.11.10.3 B.10.3 Absorbing Posterior: Numerical Example Let $T = 4$, $\alpha_t = 1 - t/T$, so $\alpha_0 = 1$, $\alpha_1 = 0.75$, $\alpha_2 = 0.5$, $\alpha_3 = 0.25$, $\alpha_4 = 0$.

Given $x_0^i =$ ”cat” and $x_3^i = \mathtt{M}$ (masked at step 3):

$$\begin{aligned}q(x_2^i \mid x_3^i = \mathtt{M},\, x_0^i = \text{"cat"}) &= \frac{\alpha_2 - \alpha_3}{1 - \alpha_3}\,\mathbf{e}_{\text{"cat"}} + \frac{1 - \alpha_2}{1 - \alpha_3}\,\mathbf{e}_{\mathtt{M}} \\ &= \frac{0.5 - 0.25}{1 - 0.25}\,\mathbf{e}_{\text{"cat"}} + \frac{1 - 0.5}{1 - 0.25}\,\mathbf{e}_{\mathtt{M}} = \frac{1}{3}\,\mathbf{e}_{\text{"cat"}} + \frac{2}{3}\,\mathbf{e}_{\mathtt{M}}\end{aligned}$$

At step $t = 3$, a masked token at step $t = 2$: 1/3 chance it was unmasked (= “cat”), 2/3 chance it was still masked. The model must predict “cat” with confidence proportional to this posterior.

9.11.10.4 B.10.4 Entropy of Generated Text Generated sequence entropy $H = 5.45$ nats per token (observed for remdm-loop at T=1000). Character perplexity: $e^{5.45} = 233$.

If entropy drops to $H = 5.35$ nats (remdm-conf at T=1000, a drop of 0.10 nats): - Character perplexity ratio: $e^{0.10} = 1.105$ - Interpretation: effective vocabulary 10.5% smaller — not catastrophic, but a consistent signal of mode-seeking behaviour.

The corresponding gen-ppl change is **larger** than this entropy change alone would suggest, because mode-seeking also reduces $D_{\mathrm{KL}}(p_\theta\|q_{\text{GPT-2}})$: a less diverse model is “closer” to GPT-2’s distribution in the forward KL sense.

9.11.11 B.11 Quick-Reference: Key Inequalities

Inequality	Statement	Used for
Jensen	$f(\mathbb{E}[X]) \le \mathbb{E}[f(X)]$ for convex f	ELBO derivation
Gibbs	$D_{\mathrm{KL}}(P \Vert Q) \ge 0$	All KL-based bounds
Data processing	$I(X;Z) \le I(X;Y)$ if $X \to Y \to Z$	Forward process information loss
Chain rule	$H(X_1, \dots, X_n) = \sum_i H(X_i \mid X_{1:i-1})$	Autoregressive decomposition
Conditioning	$H(X \mid Y) \le H(X)$	Information profile optimality
Log-sum	$\sum_i a_i \ln(a_i/b_i) \ge (\sum_i a_i) \ln(\sum_i a_i / \sum_i b_i)$	Rate-distortion bounds

---

9.11.12 B.12 Notation Conventions Used Throughout This Document

Symbol	Meaning
$\mathbf{x} = (x^1, \dots, x^L)$	Token sequence of length L
$\mathbf{x}^{\ i}$	All tokens except position i
$\alpha_t \in [0,1]$	Survival probability at step t (prob. of being unmasked)
`M` or `[MASK]`	Mask token
$q(\cdot)$	Forward process (fixed, data-independent)
$p_\theta(\cdot)$	Reverse (generative) model with parameters θ
$D_{\mathrm{KL}}(P \Vert Q)$	KL divergence from Q to P
$H(P,Q)$	Cross-entropy of P under Q
$H(X)$	Entropy of X
$I(X;Y)$	Mutual information
$I^i(\mathbf{x})$	Information profile at position i: $H(x^i \mid \mathbf{x}^{\ i})$
E_{fact}	Total factorisation error $= \sum_t D_{\mathrm{KL}}(\text{true} \Vert \text{factored})$
ELBO	Evidence lower bound $= \log p(\mathbf{x}) - D_{\mathrm{KL}}(q \Vert p_\theta(\cdot \mid \mathbf{x}))$
Δ_{V-1}	Probability simplex over V tokens
τ	Temperature parameter (softmax scaling)
π	Probability vector in Δ_{V-1}

9.12 Appendix B: Repository Index

Repository	GitHub URL	Submodule path	Role in thesis
kuleshov-group/remdm	https://github.com/kuleshov-group/remdm	`external/remdm`	ReMDM strategies + MDLM-OWT inference
SeunggeunKimkr/PRISM	https://github.com/SeunggeunKimkr/PRISM	`external/PRISM`	PRISM quality head (Sudoku + OWT)
maple-research-lab/RemeDi	https://github.com/maple-research-lab/RemeDi	`external/remedi`	RemeDi-RL, RemeDi-Instruct inference
kuleshov-group/mdlm	https://github.com/kuleshov-group/mdlm	`external/mdlm`	Original MDLM — independent baseline
louaaron/Score-Entropy-Discrete-Diffusion	https://github.com/louaaron/Score-Entropy-Discrete-Diffusion	`external/sedd`	SEDD PC-sampler baseline
GSAI-ML/LLaDA	https://github.com/GSAI-ML/LLaDA	HF only	LLaDA-8B — loaded via `transformers`

Compiled with:

```
pandoc comparison.md -o comparison.pdf \
  --pdf-engine=xelatex -V geometry:margin=2.5cm \
  --toc --toc-depth=2 --number-sections
```